Cloud Removal With SAR-Optical Data Fusion Using a Unified Spatial–Spectral Residual Network

Yuxi Wang, Bing Zhang, *Fellow, IEEE*, Wenjuan Zhang, *Member, IEEE*, Danfeng Hong, *Senior Member, IEEE*, Bin Zhao, *Member, IEEE*, and Zhen Li

***Abstract*— Cloud contamination greatly limits the potential utilization of optical images for geoscience applications. An effective alternative is to extract data from synthetic aperture radar (SAR) images to remove clouds due to the strong penetration ability of microwaves. In this article, we propose a novel unified spatial–spectral residual network that utilizes SAR images as auxiliary data to remove clouds from optical images. The method can better establish the relationship between SAR and optical images and be divided into two modules: feature extraction and fusion module and reconstruction module. In the feature extraction and fusion module, a gated convolutional layer is introduced to discriminate cloud pixels from clean pixels, which makes up for the lack of distinguishing ability of vanilla convolutional layers and avoids the error of cloud areas in feature extraction. In the reconstruction module, spatial and channel attention mechanisms are introduced to obtain global spatial and spectral information. The network is tested on three datasets with different spatial resolutions and compositions of land covers to verify the effectiveness and applicability of the method. The results show that the method outperforms other mainstream algorithms that simultaneously use SAR images as auxiliary data with a gain of about 2.3 dB in terms of peak signal-to-noise ratio PSNR on the SEN12MS-CR dataset.**

***Index Terms*— Cloud removal, data fusion, deep learning, residual network, synthetic aperture radar (SAR)-optical.**

Manuscript received 24 May 2023; revised 5 November 2023; accepted 12 November 2023. Date of publication 4 December 2023; date of current version 15 December 2023. This work was supported in part by the Youth Innovation Promotion Association, CAS; in part by the Defense Industrial Technology Development Program; in part by the National Natural Science Foundation of China (NSFC) under Grant 42201503; and in part by the National Key Research and Development Program of China under Grant 2022YFB3903401. *(Corresponding authors: Bing Zhang; Wenjuan Zhang.)*

Yuxi Wang is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, also with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: wangyuxi20@mails.ucas.ac.cn).

Bing Zhang is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zb@radi.ac.cn).

Wenjuan Zhang is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhangwj@aircas.ac.cn).

Danfeng Hong is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: hongdf@aircas.ac.cn).

Bin Zhao is with the School of Information Science and Engineering, Shandong Agricultural University, Tai'an 271002, China, and also with the Faculty of Electrical and Computer Engineering, University of Iceland, 102 Reykjavik, Iceland (e-mail: haobin@radi.ac.cn).

Zhen Li is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: lizhen02@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2023.3339210

I. Introduction

NUMEROUS remote sensing data have been acquired and have been used for geoscience applications to facilitate the monitoring of the land surface environment [1]. However, the optical imagery acquired by satellite sensors is inevitably contaminated by thick clouds and their shadows, which enormously increases the difficulty of further applications. The global cloud amount provided by the International Satellite Cloud Climate Program (ISCCP) shows that 63% of the Earth is covered by clouds [2]. Cloud cover is a common problem in optical satellite imagery, which will lead to large quantities of missing information and cause great difficulties in subsequent image interpretation and application. Thus, reconstructing cloud-free images through cloud removal methods is much sought after.

Generally, cloud removal is a process of missing information reconstruction. Numerous methods have been proposed for reconstructing missing data for cloud-covered images. From the perspective of the complementary information source, existing methods for thick cloud removal can be roughly divided into three major categories, spatial-based approaches, multitemporal-based approaches, and multisource-based approaches [3].

The spatial-based approaches are the most basic and classic cloud removal methods. They are based on the spatial autocorrelations to recover the missing information. The spatial-based methods include interpolation-based methods [4], [5], spreading or diffusion based methods [6], [7], exemplar-based methods [8], [9], and total variational methods [10], [11], [12]. Generally, they are effective for removing the small cloud with regular texture [13]. However, there is no reference image, the spatial-based methods perform unsatisfactory when it comes to large cloudy regions and cannot reconstruct the high-frequency texture information.

Since the images obtained from the same sensor have the same system characteristics. Thus, the adjacent time images can be used for the temporal-based methods as auxiliary data to restore the information of the missing area. The temporal-based methods usually establish a linear relationship between cloud-contaminated images and reference ones, Chen et al. [14] put forward the neighborhood similar pixel interpolator (NSPI) for the Landsat enhanced thematic mapper plus (ETM+) scan-line corrector (SLC) malfunction problem. Then, the modified NSPI algorithm (MNSPI) was proposed [15] for removing thick-cloud contamination which required

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

cloud-free images as auxiliary images. Chen et al. [16] developed the spatial and temporal weighted linear regression (STWR) model and made full use of the effective information in the cloud-free area to remove thick clouds in the Lansat-8 operational land imager (OLI) image. Li et al. [17] exploited complementary information in adjacent temporal satellite images to replace cloud-contaminated regions after precise radiometric correction. Compared with the spatial-based approaches mentioned earlier, the temporal-based methods are less influenced by the spatial size and the landscape heterogeneity of the cloudy areas, and they can yield a favorable cloud-free result. However, the temporal-based methods cannot handle the case that cloud-free reference images are not available in a short period of time. Especially, for some optical satellites with high spatial resolution, a short revisit period cannot be guaranteed.

Multisource-based methods take advantage of data obtained by other sensors to remove the cloud. Shen et al. [18] developed a cloud removal method based on multisource data fusion, which employed a cloud-free Landsat image as reference and introduced two auxiliary moderate resolution imaging spectroradiometer (MODIS) images with close acquisition dates to the contaminated Landsat images. Li et al. [19] proposed the spatial and temporal nonlocal filter-based data fusion model (STNLFFM) to generate images by fusing a reference Landsat image and two or more MODIS images. Then, in order to retain the cloud-free information, the fused reference image was used to remove the cloud of the cloud image based on a nonnegative matrix factorization (NMF) and error correction. Using images obtained by other optical sensors with higher temporal resolution as auxiliary information to shorten the time interval between cloud-contaminated images and cloudless images. However, a higher temporal resolution source usually has lower spatial resolution [19], which is not conducive to the recovery of missing images. Besides, as the reference image is an optical image, which will be easily influenced by clouds, it is difficult to obtain cloud-free optical reference images for areas covered by clouds for a long time.

Also, there is a series of work to remove clouds based on synthetic aperture radar (SAR) sources as reference [20]. Due to the strong penetration ability of microwaves to clouds, it can work in all weather and at all times. Even under thick cloud coverage, compared with optical imagery, it can reflect real-time ground information. And a series of high-spatial-resolution SAR images became available in recent years from Advanced Land Observing Satellite (ALOS) phased array-type L-band SAR (PALSAR, L-band) [21], TerraSAR-X [21], Gaofen-3 [22], and airborne SAR. Therefore, it has great advantages to use SAR data as a reference to achieve cloud removal from remote sensing images. For example, Chen et al. [23] developed a novel deep Siamese convolutional multiple-layers recurrent neural network (RNN) (SiamCRNN) to utilize multisource data. SiamCRNN is able to extract spatial–spectral features from homogeneous or heterogeneous image patches. To improve the accuracy, Zhu et al. [24] proposed an inpainting-based cloud removal method, which took cloud-free patches from the contaminated image and introduced SAR data as the structures of missing regions for restoration. The priority of each cloudy patch is decided by the distance to cloudless patches and the complexity of structure information. Generative adversarial networks (GANs) have been introduced to produce cloud-free images from SAR images [25], [26], [27], [28], [29], [30], [31], [32], [33]. In [25], the cycle GAN (CycleGAN) [26] was introduced and trained using unpaired SAR images and optical images to obtain simulated optical data to replace the corrupted data. Bermudez et al. [27], [28] used a conditional adversarial network (cGAN) [29] to obtain cloud-free data to map the relationship between pairwise SAR and optical images. Grohnfeldt et al. [30] used SAR data and corrupted optical imagery based on a cGAN to obtain cloud-free Sentinel-2 imagery. Due to the problems of speckle noise and lack of color information in SAR images, Darbaghshahi et al. [31] presented two GANs to remove the cloud. The first translated SAR images to optical images and the second removed clouds using the translated images of prior GAN, which increased the quality of output images. To improve the interpretation of SAR images, Li et al. [32] adapted a modified cGAN-based (McGAN) method based on structure similarity index measure (SSIM) and L1 norm with the aim to generate synthetic cloudless images. In [34], the style-based GAN architecture (StyleGAN) was proposed which directly controlled the intensity of image features at different scales and led to an automatic learned of high-level attributes and random variation in the generated cloudless images. Considering the characteristic artifacts in images generated by StyleGAN, StyleGAN2 was presented [35] to eliminate artifacts, which redesigned the normalization used in the generator and changed the training approach. Karras et al. [36] adapted a modified StyleGAN (StyleGAN3) to solve the problem of the dependence of features on absolute coordinates, realize the true invariance of image translation and rotation, and greatly improve the quality of cloudless synthetic images. As GANs may suffer from training and prediction instabilities when fed with input data with large cloud cover, a deep residual convolutional neural network (CNN) architecture [37] was proposed to reconstruct Sentinel-2 images using SAR data as auxiliary data. The model used the residual net (ResNet) [38] architecture as the backbone that was robust to the presence of large thick clouds in the optical images and enabled the reproduction of cloud-free areas and the reconstruction of cloudy areas. For large areas of thick cloud and cloud shadow cover, Zhang et al. [39] proposed a novel spatiotemporal patch group deep learning framework through multitemporal cloudy images. Besides, weighted aggregation and progressive iteration were utilized for reconstructing the results. A series of experiments are performed on single/multitemporal Sentinel-2 and Landsat-8 images to validate the effectiveness. Gao et al. [40] combined the advantages of multitemporal optical images and SAR images in a CNN to remove the cloud. To address the modal heterogeneity of multimodal images, Chen et al. [41] presented an unsupervised framework based on two structural relationships of multimodal data and graph convolutional networks to learn robust and representative features. The cloudless optical images generated by these methods usually use vanilla

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

convolution (VC), which introduced errors in cloudy areas when extracting features and was not conducive to the utilization of nonneighborhood spatial and spectral information.

In general, the acquisition time interval between cloud images and multisource reference images is much shorter than that of multitemporal reference images. Since the electromagnetic waves of SAR can penetrate clouds and collect data in most weather conditions. SAR images have advantages for cloud removal compared with optical images acquired by other sensors. The reliability of this information is higher than any other references.

Although the above-mentioned methods show excellent performance in image inpainting tasks, they still have some shortcomings. First, the cloud removal networks designed earlier [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41] not discriminate between cloud and cloudless regions and used vanilla convolutional layers to extract features. VC calculates each pixel as a valid value and is suitable for classification and detection tasks, in which all pixels of the input image are valid. Therefore, all pixels are used for computation and the invalid values of cloud regions will introduce errors for subsequent feature extraction. Especially in the face of the lack of large areas of irregular clouds, a discriminative mechanism considering their unique characteristics can improve cloud removal performance. Second, these methods usually use a series of convolutional layers to extract features in a sliding window manner. The operation obtains the local features at that location in the local receptive field provided by the window. Therefore, it is difficult to build a nonneighborhood correlation and spectral correlations model only by using the convolutional network. These networks lack adequate utilization of nonneighborhood spatial and spectral information.

In order to solve the problems earlier, we developed an SAR-optical data fusion with a unified spatial–spectral residual network for cloud removal (USSRN-CR). The main contributions of this work are summarized as follows:

1) A novel deep learning-based method is developed for cloud removal and the cloud-free optical image is estimated directly from multisource reference images through the gate convolution (GC). The GC can distinguish and filter out invalid pixels and only perform feature extraction on valid pixels for subsequent image inpainting, which solves the problem that VC for deep learning-based methods unavoidably uses information from invalid pixels that would harm the image repairing process.

2) The attention mechanism can provide a global receptive field and obtain global-scale features based on the correlation between the positions. Therefore, we introduce it into the network and combine the spatial and channel attention mechanisms to take advantage of the neighborhood and nonneighborhood spatial information in images and model spectral interdependencies. The outputs of these two attention modules are fused to obtain key features with global information to recover more details of irregular cloud regions.

3) To validate the feasibility and applicability of the proposed method, a series of simulated and real experiments are performed on GF-2/3, airborne SAR/optical data, and sentinel-1/2, which cover different spatial scales, different remote sensors, and different land cover. The results demonstrate that the proposed method is robust and can effectively remove thick clouds and cloud shadows. In addition, compared with other algorithms, the proposed method shows superiority in both qualitative and quantitative results. The code of USSRN-CR is available at https://github.com/WenjuanZhang-aircas/USSRN-CR.

The rest of this article is organized as follows. Section II presents the related works. In Section III, the workflow for the proposed method is presented in detail. In Section IV, some simulation and real-data experiments are conducted to verify the effectiveness and flexibility of the proposed method in spaceborne GF-2/3, airborne SAR/optical and sentinel-1/2 data. The conclusion is summarized in Section V.

II. Related Works

A. Residual Net Architecture

Theoretically, the deep network with more hidden layers has strong feature learning capability and feature representation. However, it also brings about the problem of gradient vanishing and gradient exploding [42]. The popular ResNet was first proposed to figure out this problem and it has been proven to be one of the most successful network structures in computer vision.

The ResNet architecture makes use of skip connections to shuttle the information to lower parts of the network. It is divided into two parts, the direct mapping part and the residual part. The skip connection performs an additive identity mapping, i.e., the input direct mapping part is added to the output of the residual part. To further understand the principle, we suppose x is the original input and $G(x)$ is the output of the whole ResNet network. By adding the additive skip connection, the output of the residual part $F(x)$ can be represented as $G(x) - x$.

Nowadays, some typical cloud removal methods [37], [43] in remote sensing images based on ResNet have been proposed. In [37], the ResNet architecture is used as a backbone to reconstruct Sentinel-2 images using SAR data as auxiliary data for the task of multisource cloud removal, which is an effective extension to the image reconstruction domain. The model is robust to the presence of large thick clouds in the optical images and enables the reproduction of cloud-free areas and the reconstruction of cloudy areas. Analogously, to maintain global consistency in optical images and handle speckle noise in SAR images, Fang et al. [43] proposed a novel global–local fusion-based cloud removal (GLF-CR) algorithm that used the ResNet to leverage the complementary information embedded in SAR images.

In general, the ResNet architecture provides a direct path to transfer valid information in clear areas to the output. The entire network learns the predicted corrections which will be minor for the clear sky and larger for thick clouds. Therefore, it enables adequate reproduction in cloud-free areas and better reconstructs cloudy areas. In addition, ResNet can reduce the vanishing gradient problem and optimize large models in a relatively fast manner to yield good performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

B. Gate Convolution

When facing a large area of information loss in the image, the standard convolution network is usually filled based on the average value of effective pixels and missing areas, which is easy to cause problems, such as color difference and blur, affecting the visual effect.

To deal with the above-mentioned problems, Liu et al. [44] proposed partial convolution, which conducted a binary mask corresponding to the damaged area to constrain the conditioned region. Features obtained in the visible area are retained through constant updating of sufficient layers. However, the hard mask will be automatically updated according to a simple rule, that is, as long as there is a valid value, the location in the mask will be updated to be valid. As the convolutional network deepens, invalid pixels will gradually penetrate into effective pixels. To cope with this limitation, GC [45] utilized two parallel convolutional pathways on feature and mask, respectively, and automatically learned a soft mask to discriminate the valid and invalid pixels. Dai et al. [46] proposed a novel cloud removal method based on a gated convolutional network that reconstructed invalid pixels with the auxiliary information from multitemporal images. It can better solve the problem of treating all inputs as legal pixels in the VC and provide a learnable dynamic feature selection mechanism for each spatial location. Therefore, in order to deal with the problem of invalid pixels on the optical image, we establish a multibranch structure in the proposed model to extract the features of SAR images and optical images, respectively, and introduce a series of GC in the optical branch to extract the features of cloud-free regions on the cloud image.

C. Attention Module

Generally, the standard CNNs are limited by the local and short-range dependence information [47]. To overcome the disadvantages, an attention mechanism is proposed to capture long-range dependencies. It stems from human intuition and is introduced into natural language processing (NLP) for machine translation tasks. The Google Brain team [48] gives a general definition of the attention mechanism. Given a query and a key-value pair, the process involves mapping the query to the correct input. The attention is abstracted as weighting of the value which is calculated based on the similarity between the query and key. In recent years, attention mechanisms combined with deep learning models have become increasingly popular. Several methods have been proposed in semantic segmentation [49], object detection [50], and scene classification [51].

Besides, attention is important to learn patterns crossing spatial locations in image inpainting [52], [53], [54], [55], [56], [57]. Yu et al. [52] designed a two-stage GAN architecture from coarse to fine and introduced our baseline model with a novel spatial attention module, which significantly improved image inpainting results. Xu et al. [53] introduced the spatial attention mechanism-based GANs for cloud removal (AMGAN-CR) method for Landsat images. However, attention modules are expensive to compute, and efficient attention (EA) [54] was proposed with substantially less memory and computational costs. Also, the proposed method [55] introduced a channel attention mechanism into the residual learning (RL) path, which suppresses thin clouds and enhances the details of ground scenes simultaneously. In our model, in addition to using an individual spatial or channel attention mechanism in CNNs, the combination of channel and spatial attention is considered in our network to learn global spatial and spectral information, which is beneficial for cloud removal.

III. Methodology

A. Overview of the Cloud Removal Network

This article proposed an architecture to reconstruct the missing parts of optical images with SAR images as additional data based on the unified spatial–spectral residual network. An SAR image of the same scene as a form of prior and input the cloudy optical image into the network to complete the estimation from cloudy to cloudless images. And cloud-free images of the same scene are presented to the network as the target for loss computation.

The overall workflow of our method is shown in Fig. 1. It cast missing reconstruction into two collaborative subtasks, i.e., feature extraction and fusion module and reconstruction module. Besides, the input image will be shuttled to an additional layer right over a long skip connection before the final output to transfer effective information in clear areas. The multisource input images are extracted with the multiscale features, respectively, and fused by the feature extraction and fusion module. It contains three branches. For the two main branches, every branch has two layers which are the convolution layer and multiscale features blocks (MFBs). Although optical images and SAR images are quite different due to different imaging mechanisms, their edge structural information is very similar in the same scene [58]. Also, some inpainting tasks based on deep learning take canny edges as the important prior [59], [60]. Fig. 2(a) and (b) are the optical images and SAR images of the same scene. The corresponding edge maps extracted by the canny operator are shown in Fig. 2(c) and (d), and the contour lines of roads and cities or farmlands are generally similar. Thus, we use the canny detector to extract the line and edge maps of the SAR image to obtain structural information following a convolution layer in the third branch. Then, we need to concatenate the multifeature of SAR and optical images and the edge feature according to a certain ratio determined by experiments.

In the reconstruction module, the fused features are reconstructed as cloud-free predicted images. The first two layers, respectively, contain a convolution layer with subsequent ReLU activation, following five residual blocks (RB). After a spatial and channel attention block (SCAB), the network ends with five RB and the final convolution layer restores the spectral dimension to the number of bands in the optical image. The RB contains five layers and a residual connection. These five layers include a 2-D convolution layer with subsequent ReLU activation, a second 2-D convolution layer with ReLU activation, and a final convolution layer. For all convolutional layers, the size of the kernel is 3×3.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

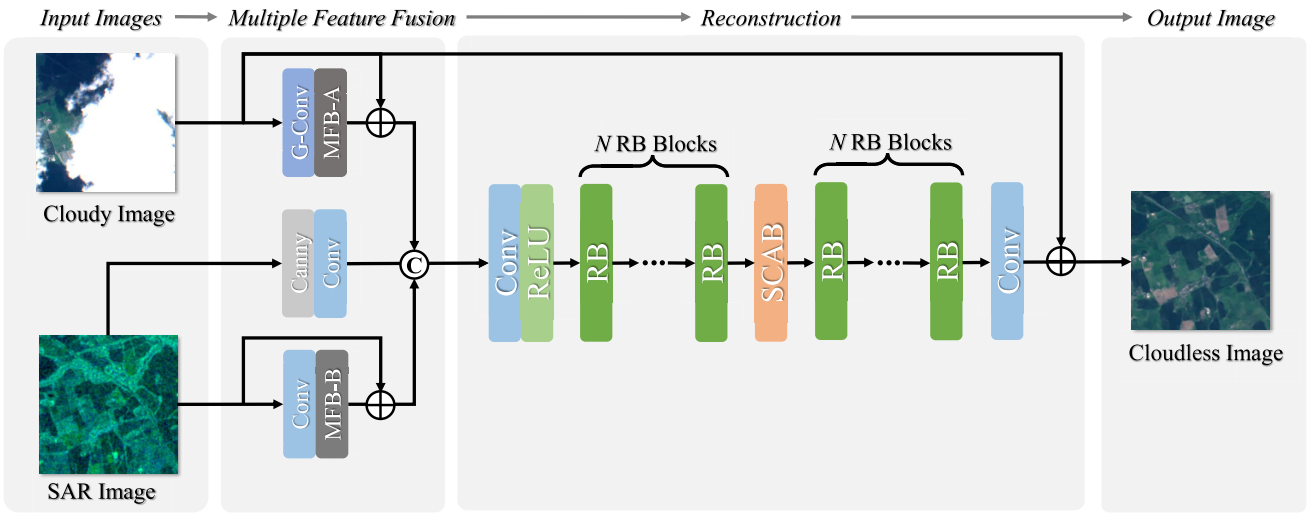


Fig. 1. Structure of the USSDRN-CR model.

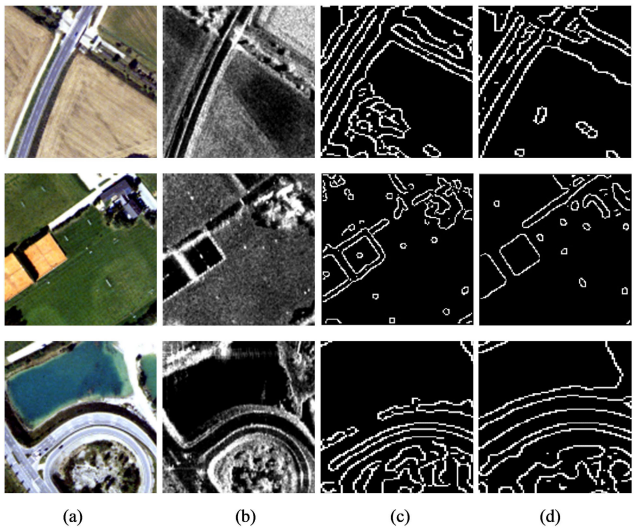


Fig. 2. SAR image and optical image of the same scene. (a) Optical image. (b) SAR images. (c) Edge map of optical image. (d) Edge map of SAR image.

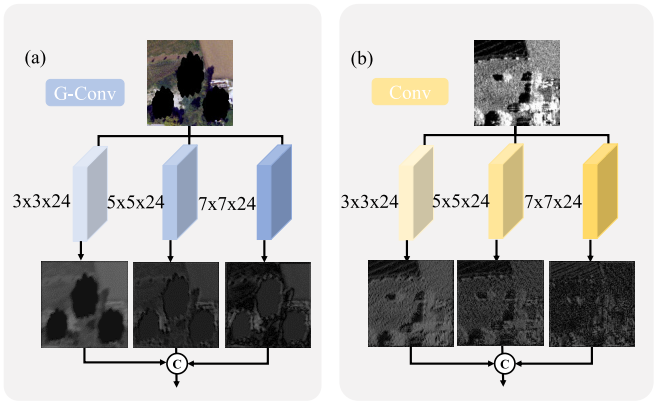


Fig. 3. Multiscale convolution feature extraction block. (a) MFB-A. (b) MFB-B.

Considering our own experiments and the experience reported by Meraner et al. [37], the amount of our architecture feature maps in RB is 256.

B. Feature Extraction and Fusion Module

The SAR and optical images are extracted with the multiscale features, respectively, and fused in the feature extraction and fusion module. To extract multiscale features, we introduce the MFB into two main branches. Also, we use the GC to replace the VC to suppress invalid values on optical cloud images. Detailed descriptions of each component of the module are provided in the following.

1) Multiscale Features Blocks: Remote sensing images contain ground objects of various scales, and the reconstruction of missing images depends on contexts of different scales. It is very important to obtain features of various scales and use them for cloud removal.

There are two types of MFB for optical and SAR images, namely, MFB-A and MFB-B, respectively. For optic images, as shown in Fig. 3(a), we use MFB-A to extract features and symbols G-Conv denote GC. MFB-A contains three GC operations of 3×3, 5×5, and 7×7 kernel sizes and produces feature maps of 24 channels. As shown in Fig. 3(b), MFB-B is conducted to extract features and Conv operation means VC. MFB-B also contains three VC operations of 3×3, 5×5, and 7×7 kernel sizes and generates an output of 24 feature maps, respectively. All three convolutions are conducted on the SAR images. These feature maps with different scales of contextual information are then concatenated together.

2) Gate Convolution: A VC functions equally on each pixel of an image which uses filters of the same shape but different coefficients to extract local features. However, in cloud removal tasks, there are numerous invalid pixels in the corrupted optical image. The features affected by invalid pixels are gradually synthesized into deeper layers and the invalid pixel will gradually become a valid pixel in a deep learning network, which inevitably impacts the restoring results such as color discrepancy and blurriness. Hence, valid and invalid

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

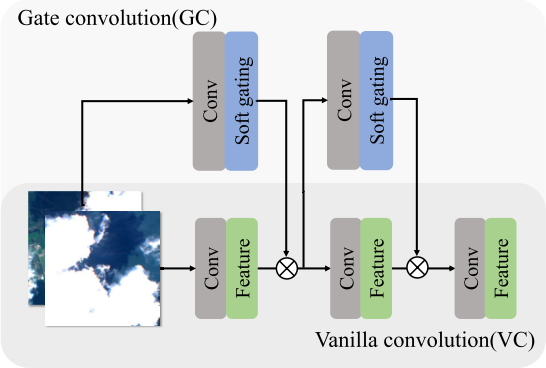


Fig. 4. GC and VC.

pixels should be dealt with separately. We propose the GC to replace VC to deal with this problem.

As shown in Fig. 4, GC establishes a learnable dynamic feature selection mechanism at different positions of the feature layer, learns the optimal mask through the network, and assigns the soft-coded value to each spatial position to discriminate valid and invalid pixels. It is formulated as

$$\text{Gating}(y,x) = \sum\sum W_g \times I + b_g \tag{1}$$

$$\text{Feature}(y,x) = \sum\sum W_f \times I + b_f \tag{2}$$

$$O(y,x) = \Phi(\text{Feature}(y,x)) \odot \sigma(\text{Gating}(y,x)) \tag{3}$$

where W_g and W_f are the weights of convolution operations, b_g and b_f are the corresponding biases. Φ is the ReLU [61] activation function. σ is the sigmoid activation function to make the feature gated between 0 and 1. The multiplication of feature $\Phi(F)$ and the learning mask $\sigma(G)$ is the gated convolution.

In summary, GC employs a learnable mask to make the network learn the distinction between valid and invalid values and make full use of the information in the effective value area which is beneficial to generate cloud-free images.

C. Reconstruction Module

The cloud-free predicted images are finally obtained in the reconstruction module. Considering the long-range spatial and spectral correlation, the attention block is introduced to make full use of spatial–spectral information as prior information to reconstruct cloud-free images.

Specifically, since CNN mainly deals with the information between local regions, it is difficult to capture feature dependencies in the spatial and spectral dimensions, respectively. Therefore, these methods lack the utilization of global spatial information and spectral information. While attention modules can better utilize the global feature information, which is important for image inpainting [52], [53], [54], [55], [56], [57]. It provides a global receptive field, which learns the relationship between a certain position and all other positions. The pixel of each position not only depends on other positions adjacent to this position but also introduces positions that are farther away and have a greater amount of information. Hence, the attention block is introduced into the reconstruction block, and the global information of the image is obtained through SCAB.

As illustrated in Fig. 5, the SCAB includes the spatial attention block (SAB) and the CAB. The SAB utilizes neighbor and nonneighbor information in the image. The attention mechanism is used to capture the spatial dependencies between any two locations of the feature maps. The feature is updated by a weighted summation of features at all locations, and the weights are calculated by the feature similarities between any two positions. Therefore, both neighbor and nonneighbor regions with similar characteristics can promote each other. The CAB can take full advantage of long-range information and capture the channel dependencies between any two-channel maps. Like the SAB, we also use the attention mechanism to capture the spectral dependencies. Finally, we aggregate the outputs of the two attention modules to obtain better feature representations for cloud removal.

1) Spatial Attention Block: New features are generated with global spatial information through the following three steps. The first step is to obtain a spatial attention map that models the spatial relationship between any two positions of the features. Next, we multiply the attention matrix and the original feature. Finally, we sum up the above-mentioned result matrix and the original features. Also, we leverage the EA [54] module detailed in Fig. 5(a) to reduce significantly computational cost. Particularly, for the input feature $X \in R^H \times W \times C$, we first feed it into a convolution layer and then reshape it to $R^{\text{HW}} \times C$. The expression of the Q, K, and V is as follows:

$$Q = \text{softmax}_{\text{row}}(\text{conv}(X)) \tag{4}$$

$$K = \text{softmax}_{\text{col}}(\text{conv}(X)) \tag{5}$$

$$V = \text{conv}(X) \tag{6}$$

where $\text{softmax}_{\text{row}}$ and $\text{softmax}_{\text{col}}$ mean the softmax operations on the row and column individually. After that, we perform a matrix multiplication between the transpose of K and V to calculate the spatial attention map S, which establishes the spatial relationship between any two pixels. This operation can be written as follows:

$$S = K^T V. \tag{7}$$

A matrix multiplication is performed between Q and S, the output will be reshaped back to $R^{H\times W\times C}$. Finally, we multiply it by a scale parameter α and perform an elementwise sum operation with the features X. The process of EA E_s can be summarized as

$$E_s = \alpha \text{QS} + X \tag{8}$$

where α is initialized as 0 and gradually learns to assign more weight [62]. Since E_s is computed by $K^T V \in R^{C\times C}$, rather than standard $\text{QK}^T \in R^{\text{HW}\times\text{HW}}$. In practice, the dimension C is much smaller than HW in computer vision tasks. Furthermore, we introduce the multihead attention [48] to further reduce the dimension from to $C' = C/\text{nhead}$. It can be inferred from the above-mentioned formula that the resulting feature E at each position is a weighted sum of the features across all positions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

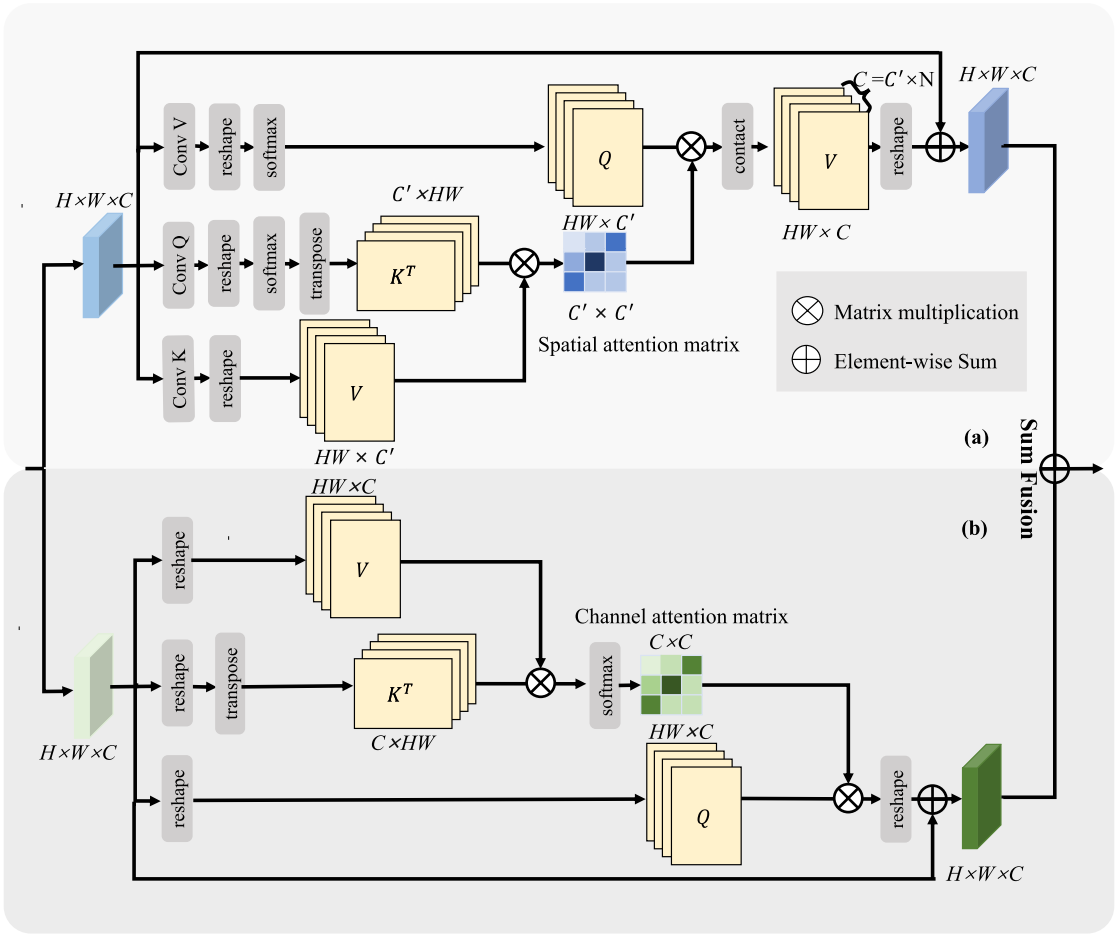


Fig. 5. Details of SCABs. (a) SAB. (b) CAB.

2) Channel Attention Block: The process is similar to the SAB and the structure is illustrated in Fig. 5(b). The CAB obtains the correlation of the spectral dimension. Also, for the input feature $X \in R^{H\times W\times C}$, we first reshape them to $R^{\mathrm{HW}\times C}$ and generate three new feature maps $Q \in R^{\mathrm{HW}\times C}$, $K \in R^{\mathrm{HW}\times C}$ and $V \in R^{\mathrm{HW}\times C}$. After that, we perform a matrix multiplication between the transpose of K and V, and apply a softmax layer to calculate the channel attention map S, which are defined as follows:

$$S = \mathrm{softmax}\left(K^T V\right). \tag{9}$$

Then, a matrix multiplication is performed between the transpose of S and Q and reshape their result to $R^{H\times W\times C}$. We multiply the result by a scale parameter β and perform an elementwise sum operation with X to obtain the final output $E \in R^{H\times W\times C}$, as follows:

$$E_c = \beta \mathrm{QS}^T + X \tag{10}$$

where β is initialized as 0. The final feature E_c of each channel is a weighted sum of the features across all channels.

We aggregate the features from these two attention blocks and perform an elementwise sum to accomplish feature fusion. The final result E is calculated as follows:

$$E = E_c + E_s. \tag{11}$$

D. Loss Function

In the proposed model, the global distribution still needs restriction to get a better result with the mean absolute error L_{global}, which is robust to large deviation. Let I_{out} represents the reconstruction result and I_{gt} is the ground truth. L_{global} can be written as

$$L_{\mathrm{global}} = \frac{1}{\mathrm{HWC}} ||I_{\mathrm{out}} - I_{\mathrm{gt}}||_1 \tag{12}$$

where HWC is the total number of pixels in all channels of the images.

The global loss function reconstructs the damaged image from a global perspective. Furthermore, the loss function should focus on the recovery of missing parts. Therefore, we utilize the cloud mask M to construct a local loss function for the missing regions. It can be summarized as follows:

$$L_{\mathrm{local}} = \frac{1}{\mathrm{HWC}} ||\mathrm{MI}_{\mathrm{out}} - \mathrm{MI}_{\mathrm{gt}}||_1. \tag{13}$$

In addition, considering human visual perception, the structural similarity SSIM loss function is selected, which represents the structural similarity between two images and can save more high-frequency information. The L_{SSIM} loss function is calculated by the following formula:

$$L_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}\left(I_{\mathrm{out}}, I_{\mathrm{gt}}\right) \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

$$\mathrm{SSIM}(I_{\mathrm{out}}, I_{\mathrm{gt}}) = \frac{(2\mu_{I_{\mathrm{out}}}\mu_{I_{\mathrm{gt}}} + C_1)(2\sigma_{I_{\mathrm{out}}I_{\mathrm{gt}}} + C_2)}{(\mu_{I_{\mathrm{out}}}{}^2 + \mu_{I_{\mathrm{gt}}}{}^2 + C_1)(\sigma_{I_{\mathrm{out}}}{}^2 + \sigma_{I_{\mathrm{gt}}}{}^2 + C_2)}. \tag{15}$$

Among them, $\mu_{I_{\mathrm{out}}}$ and $\mu_{I_{\mathrm{gt}}}$ are the mean values of recovery optical images and ground truth images, $\sigma_{I_{\mathrm{out}}}$ and $\sigma_{I_{\mathrm{gt}}}$ are the standard deviations of two images, respectively, $\sigma_{I_{\mathrm{out}}I_{\mathrm{gt}}}$ are the covariance of the two images, C_1 and C_2 are the positive constant.

The network develops a jointly global-regional loss L_{all} for the model optimization to attain holistic consistency and structural similarity. It can be expressed as

$$L_{\mathrm{all}} = L_{\mathrm{global}} + \lambda_1 L_{\mathrm{local}} + \lambda_2 L_{\mathrm{SSIM}} \tag{16}$$

where λ_1 and λ_2 as hyperparameters weighting represent the balancing thresholds for the L_{local} and the L_{SSIM}. The three loss functions are coupled with each other and constrain image inpainting in terms of content and structure information.

IV. Experimental Results and Analysis

Table I shows a detailed description of the proposed network implemented in our experiments on the airborne dataset. Filter dimensions, the number of filters, and the size of the output image are indicated. The model includes five RB before and after the SCAB as described in Section III, and one RB structure is specifically listed in Table I. To verify the feasibility and practicality of our network on the cloud removal task, a series of simulated and real experiments are implemented on different datasets with different spatial resolutions, different remote sensors, and different compositions of land covers. Details of our experiments are given as follows.

A. Datasets

We used three datasets for experiments. The Spaceborne dataset and airborne dataset are used in the simulation experiment, and the real experiment is performed on the SEN12MS-CR dataset.

For the spaceborne dataset, the data [63] from GF-2 and GF-3 with RGB bands located in Nanchang, Jiangxi, China. The GF-2 optical image was acquired on August 12, 2016, with a resolution of 3.2 m. The C-band GF-3 SAR image was acquired on August 16, 2016, and the polarized channel is VV. The land cover types are mainly farmlands and roads which are simple and less textured.

For the airborne dataset, the optical and X-band SAR data which include three bands with a size of 2813 pixels × 2289 pixels were provided in the 2001 IEEE Geoscience and Remote Sensing Symposium (GRSS) Data Fusion Competition and the spatial resolution is 1 m [64]. The polarized channel of SAR data is VV and land cover types are mainly cities, roads, and farmlands, which are relatively complex and rich in texture. The SAR images are preprocessed and denoised by the SAR block-matching 3D (SAR-BM3D) [65] algorithm.

The SEN12MS-CR dataset presented in [37], which contains triplets of cloudy and cloud-free sentinel-2 optical images and sentinel-1 C-band SAR images was created specifically for cloud removal and has a spatial resolution of 10 m. Sentinel-2 data comes from the top-of-atmosphere reflectance product and includes 13 bands. Sentinel-1 data has two polarization channels (VV and VH) as two bands obtained in the IW mode. The images are value-clipped to eliminate outliers. We randomly chose 1411 patch triplets of this dataset as the real dataset and the land cover types are mainly cities, rivers, mountains, and farmlands, which includes a variety of land cover. In order to eliminate outliers, we clipped the 13 bands of the Sentinel 2 data to be within [0, 10 000], and the two bands of the Sentinel 1 data were [−25, 0] and [−32.5, 0], respectively. Fig. 6 shows examples of patch triplets and cloud masks from the dataset.

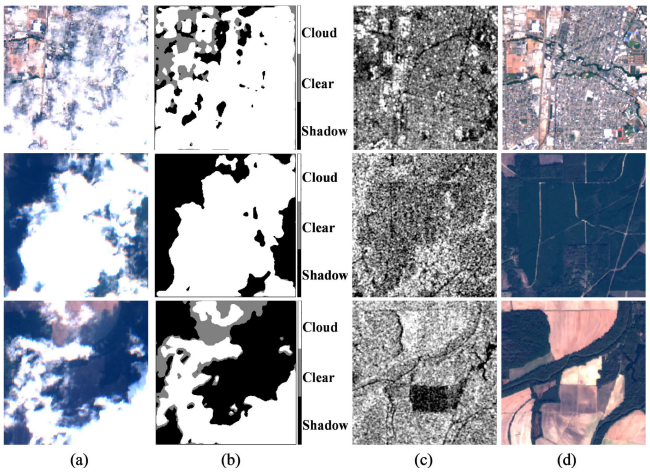


Fig. 6. Example 256 pixels × 256 pixels patch triplets from the SEN12MS-CR dataset. (a) Cloudy optical images. (b) Cloud mask. (c) SAR images (VV channel). (d) Cloud-free optical images.

The SAR and optical data are roughly registered with a deviation of fewer than two pixels and normalized to ensure stability and to keep the data within the same range before being fed into the network [37]. In Table II, the number of images for training and testing and the spatial resolution of each data are listed.

B. Experiment Details

1) Training Settings: The proposed cloud removal network is optimized via adaptive moment estimation (ADAM) [66] algorithm and the hyperparameters are set as $\beta_1 = 0.9$ and $\beta_2 = 0.999$. After a systematic search, the optimal learning rate has been found to be 7×10^{-5}. The training process is iterated 300 epochs each for our model and the other deep learning-based methods. A batch size of two samples per iteration is used for training. The weights in (16) are empirically set as $\lambda_1 = 1$, and $\lambda_2 = 1$. The algorithm is implemented using the PyTorch framework of the Windows 10 environment, with an NVIDIA GeForce RTX 3090 GPU.

2) Evaluation Indicators: The representative indicators employed to evaluate the recovery results are the peak signal-to-noise ratio (PSNR), the structural similarity index measurement (SSIM) [67], the spectral angle mapper (SAM) [68], and the correlation coefficient (CC). Images to be compared are considered as vectors $x \in R^L$ and $y \in R^L$, defined as

$$\mathrm{PSNR}(x, y) = 20 \cdot \log_{10}\left(\frac{1}{\mathrm{RMSE}(x, y)}\right) \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED CONFIGURATION OF THE PROPOSED USSRN-CR BASED ON THE AIRBORNE DATASET

Module Name		Layer Operation	Kernel Size	Filters	Output Shape
Multiple Feature Fusion	\	G-Conv	(3,3)	72	(128,128,72)
	MFB-A	G-Conv & G-Conv & G-Conv	(3,3) & (5,5) & (7,7)	24	(128,128,72)
	\	Conv & Canny	(3,3)	36	(128,128,36)
	\	Conv	(3,3)	72	(128,128,72)
	MFB-B	G-Conv & G-Conv & G-Conv	(3,3) & (5,5) & (7,7)	24	(128,128,72)
Reconstruction	\	Conv & RELU	(3,3)	256	(128,128,256)
	RB	Conv & RELU	(3,3)	256	(128,128,256)
		Conv & RELU	(3,3)	256	(128,128,256)
		Conv	(3,3)	256	(128,128,256)
	SCAB	Conv & reshape & softmax	(3,3)	64	(16384,64)
		Conv & reshape & softmax & transpose	(3,3)	64	(64,16384)
		Conv & reshape	(3,3)	64	(16384,64)
		Contact & reshape	\	256	(128,128,256)
		reshape	\	256	(16384,256)
		reshape & transpose & softmax	\	256	(256,256)
		reshape	\	256	(16384,256)
		reshape	\	256	(128,128,256)
	RB	Conv & RELU	(3,3)	256	(128,128,256)
		Conv & RELU	(3,3)	256	(128,128,256)
		Conv	(3,3)	256	(128,128,256)
	\	Conv	(3,3)	3	(128,128,3)

TABLE II
INTRODUCTION OF EXPERIMENTAL DATASETS

Dataset	Spatial Resolution(m)	Training sample	Testing sample
Spaceborne	3.2	1177	135
Airborne	1	1288	140
SEN12MS-CR	10	1131	280

$$\mathrm{SSIM}(x,y)=\frac{(2\mu_x\mu_y+\theta_1)(2\sigma_{xy}+\theta_2)}{(\mu_x+\mu_y+\theta_1)(\sigma_x+\sigma_y+\theta_2)} \tag{18}$$

$$\mathrm{SAM}(x,y)=\cos^{-1}\left(\frac{\sum_{i=1}^{L}x_i y_i}{\sqrt{\sum_{i=1}^{L}x_i^2\sum_{i}^{L}y_i^2}}\right) \tag{19}$$

$$\mathrm{CC}(x,y)=\frac{\sum_{i=1}^{L}(x_i-\mu_x)(y_i-\mu_y)}{\sqrt{\sum_{i=1}^{L}(x_i-\mu_x)^2}\sqrt{\sum_{i=1}^{L}(y_i-\mu_y)^2}}. \tag{20}$$

Images x and y are evaluated via means μ_x and μ_y, standard deviations σ_x and σ_y, covariance σ_{xy}, and constants θ_1 and θ_2. RMSE is defined as follows:

$$\mathrm{RMSE}(x,y)=\sqrt{\frac{1}{L}\sum_{i=1}^{L}(x_i-y_i)^2}. \tag{21}$$

PSNR is a commonly used pixel-level metric to quantify the reconstruction quality. SSIM is captured in terms of structure, contrast, and brightness and can measure the similarity between two images. SAM is a measure of the spectral fidelity of two images by calculating the angle between their spectra and CC is widely used to calculate the correlation. Also, we use visual evaluation to observe the reconstruction information.

3) Compared Algorithms: We selected five deep learning-based algorithms to compare with our method: the DGL model [24], the Pix2pix model [33], the McGAN model [32], the SAR-Opt-cGAN model [30], the StyleGAN3 model [36], and the DSen2-CR model [37]. These models utilize single-time SAR for cloud removal from optical images. The DGL model searches for similar patches in the cloud-free region of the contaminated image for restoration based on the structural information obtained from SAR data. The Pix2pix model and the McGAN model adopt a U-Net architecture to directly convert SAR images to optical images. The SAR-Opt-cGAN model uses a cGAN network to generate cloud-free optical data from a cloud-corrupted optic input and an auxiliary SAR image. The StyleGAN3 model adapted a modified style-based GAN architecture to acquire high-quality cloud-free synthetic images. The DSen2-CR model input is a concatenation of SAR data and damaged optical images, and a cloud-free optical image is obtained through a residual network. The above-mentioned algorithm and the proposed method are trained on the same datasets.

C. Simulated Experiment

In simulation experiments, the spaceborne dataset and the airborne dataset are cropped into 128 pixels × 128 pixels. Cloud masks are randomly rotated and cropped to generate cloud regions of different sizes and shapes and are added

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

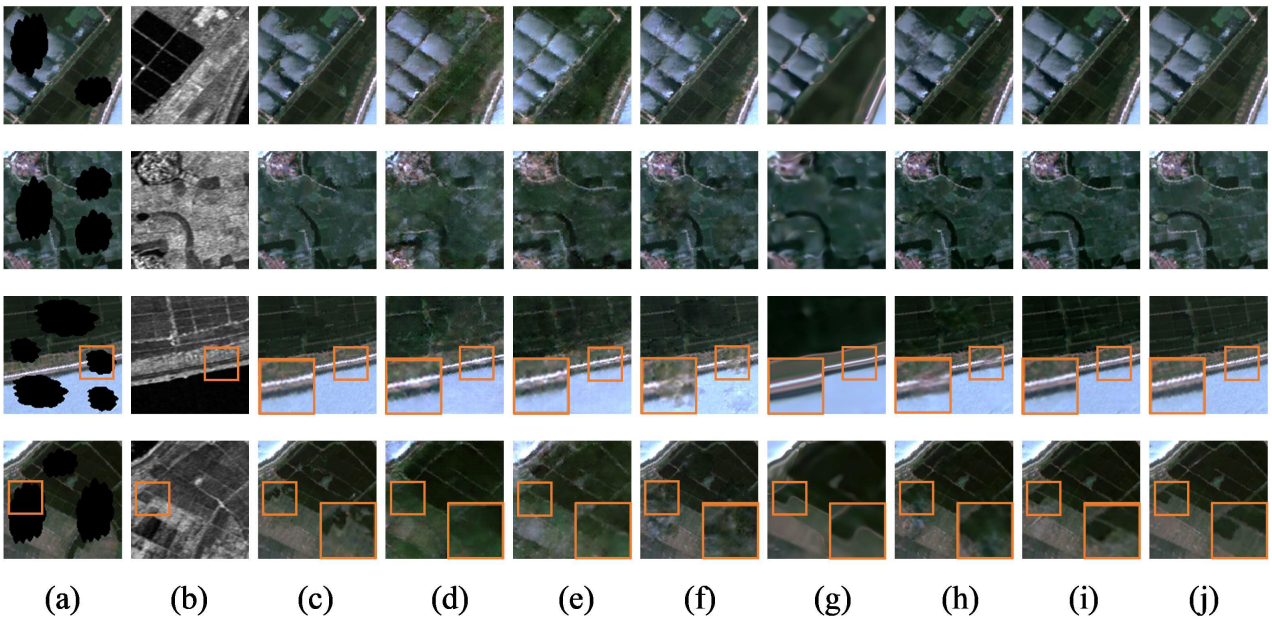


Fig. 7. Experimental results on spaceborne dataset. (a) Simulated corrupted images. (b) SAR images. (c) Results of the DGL model. (d) Results of the Pix2pix model. (e) Results of the McGAN model. (f) Results of the SAR-Opt-cGAN model. (g) Results of the StyleGAN3 model. (h) Results of the DSen2-CR model. (i) Results of the USSRN-CR model. (j) Ground truth.

to optical images to generate cloudy data. The simulated cloud area occupies 30%–40% of the optical images. Since the cloud contamination is simulated on cloudless images, the cloud images are consistent with the cloud-free region of the cloud-free images.

We use simulated cloud imagery and SAR imagery as input for training and present cloud-free optical images as the ground truth to the network as targets for loss computation. Since the cloud contamination is simulated on cloud-free images and the ground truth is reliable, we are able to evaluate the performance of different methods.

1) Results of Spaceborne Dataset: Results of simulated experiments are displayed in Fig. 7. Fig. 7(a)–(j) represents the simulated corrupted optical image, the SAR image, the results of the DGL model, the Pix2pix model, the McGAN model, the SAR-Opt-cGAN model, the StyleGAN3 model, the DSen2-CR model, our model, and the ground truth image. Besides, the quantitative evaluation results, including PSNR, SSIM, CC, SAM, and RMSE are shown in Table III.

In the spaceborne dataset, we show the restoration results of farmlands and road features. For the proposed model, the local texture and overall tone are well preserved. The reconstruction of ground object structure and texture details is superior to other models, and a better restoration effect is obtained. In the zoomed-in area, the DGL model is able to reconstruct the missing area due to the nonlocal similarity, while the restoration result shows to be less similar to the cloudless regions and the farmlands area cannot be reconstructed. The Pix2pix model and the McGAN model produce blurry farmlands and roads. The inpainting results of SAR-Opt-cGAN in cloud areas are inconsistent with the surrounding cloud-free areas. The StyleGAN3 model obtained a smooth recovery image that lacks high-frequency texture information and the DSen2-CR model cannot reconstruct the correct results. However, our model can accurately reconstruct the texture details of farmlands and the structural details of roads. The quantitative evaluation results in Table III demonstrate the results generated by the Pix2pix model, the McGAN model, and the StyleGAN3 model have large deviations in spectral information. In contrast, the DGL model and the proposed model show better performance in reconstructing the spectral information of the image.

TABLE III
QUANTITATIVE EVALUATION ON SPACEBORNE DATASET (↑: THE HIGHER SCORE INDICATES THE BETTER EFFECT ↓: THE LOWER SCORE INDICATES THE BETTER EFFECT AND THE BEST VALUE OF EACH INDICATOR IS BOLDED)

Method	PSNR↑	SSIM↑	CC↑	SAM↓
DGL	26.974	0.834	0.915	**1.169**
Pix2pix	21.902	0.576	0.852	5.475
McGAN	22.689	0.718	0.906	4.741
SAR-Opt-cGAN	23.891	0.754	0.931	3.737
StyleGAN3	24.802	0.752	0.935	4.528
DSen2-CR	25.702	0.825	0.949	2.873
USSRN-CR	**27.556**	**0.869**	**0.971**	2.654

To further measure the reconstruction quality of each model, Fig. 8 displays the scatter plots of the images in Fig. 7 for the different algorithms and their coefficient of determination (R^2). Some points of the Pix2pix model and the McGAN model deviate from the diagonal, and the points of other models are mainly distributed around the diagonal. However, our model has the highest CC and the best reconstruction performance.

2) Results of Airborne Dataset: The experimental results of airborne dataset are displayed in Fig. 9. Fig. 9(a) is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

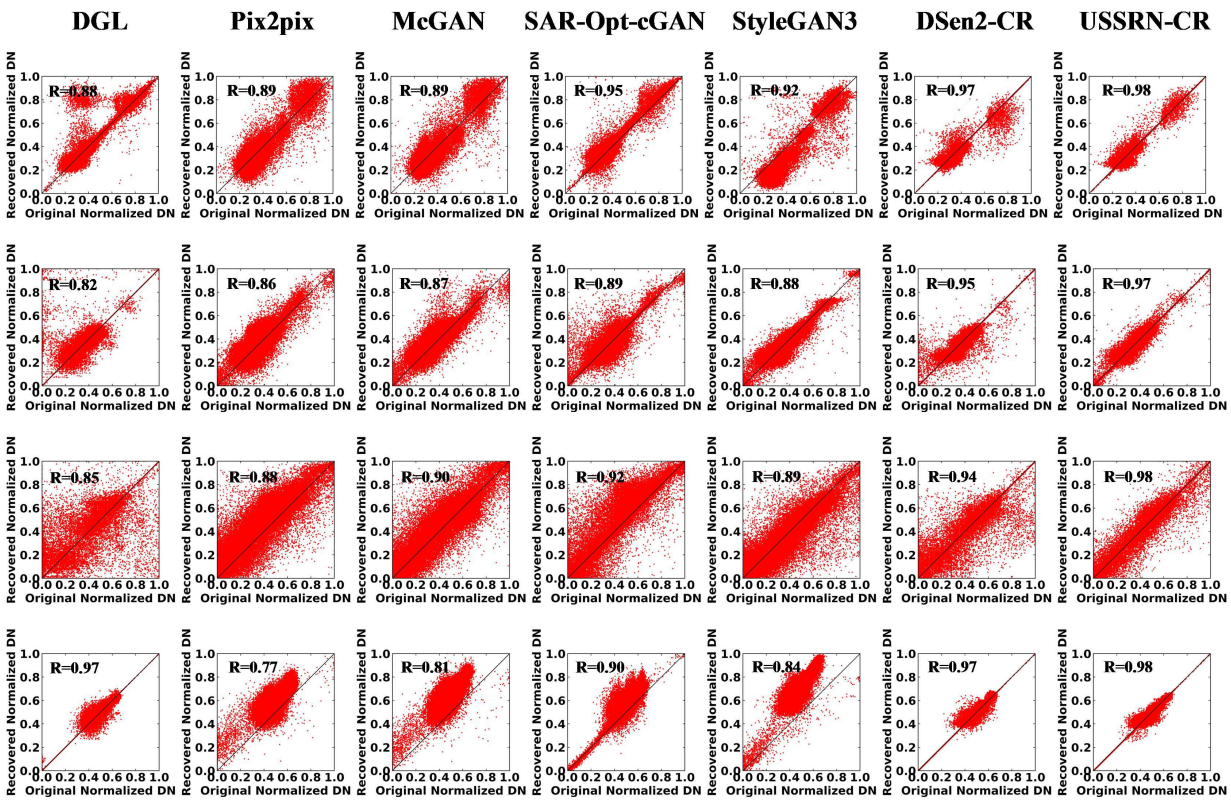


Fig. 8. Scatter diagrams between the original and reconstructed pixels in Fig. 7.

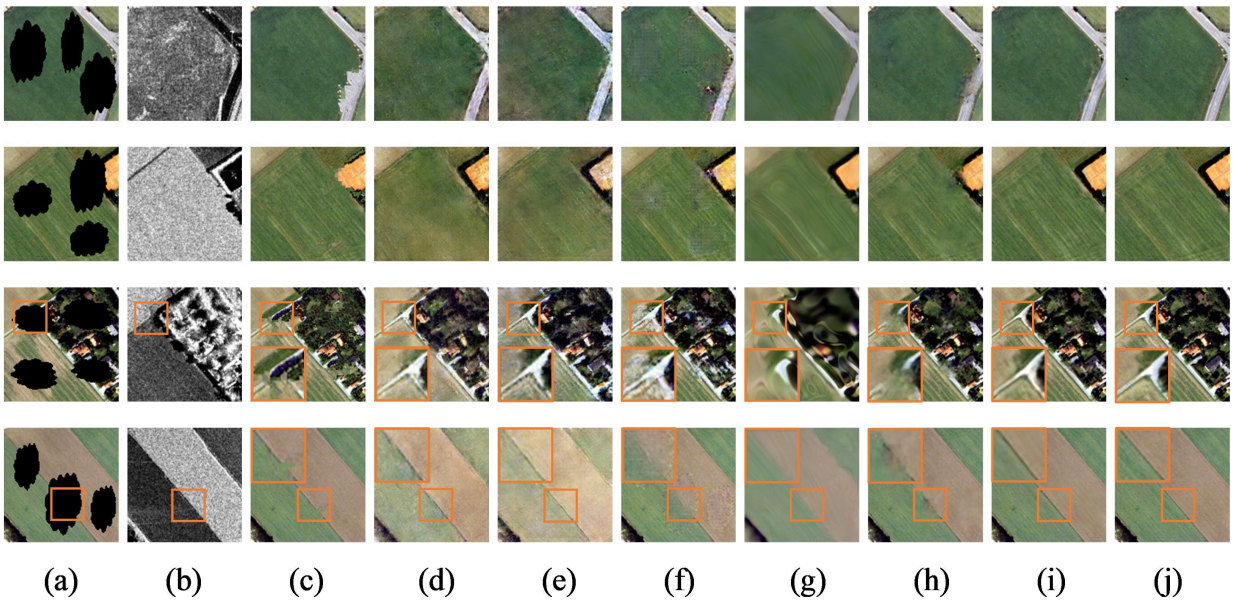


Fig. 9. Experimental results on airborne dataset. (a) Simulated corrupted images. (b) SAR images. (c) Results of the DGL model. (d) Results of the Pix2pix model. (e) Results of the McGAN model. (f) Results of the SAR-Opt-cGAN model. (g) Results of the StyleGAN3 model. (h) Results of the DSen2-CR model. (i) Results of the USSRN-CR model. (j) Ground truth.

simulated corrupted optical image. Fig. 9(b) is the input SAR image. The recovery results of the DGL model, the Pix2pix model, the McGAN model, the SAR-Opt-cGAN model, the StyleGAN3 model, the DSen2-CR model, and our model are shown in Fig. 9(c)–(i). Fig. 9(j) is the ground truth image. The scatter diagrams between the original pixels and the reconstructed pixels by different methods and their R^2 also verify the high reconstruction accuracy of the proposed framework, as presented in Fig. 10.

Table IV shows the quantitative evaluation results of different methods on the simulated airborne dataset. Our model outperforms the above-mentioned five models on all metrics. As shown in Fig. 9, we show recovery images of various objects, such as cities, roads, and farmlands. It can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

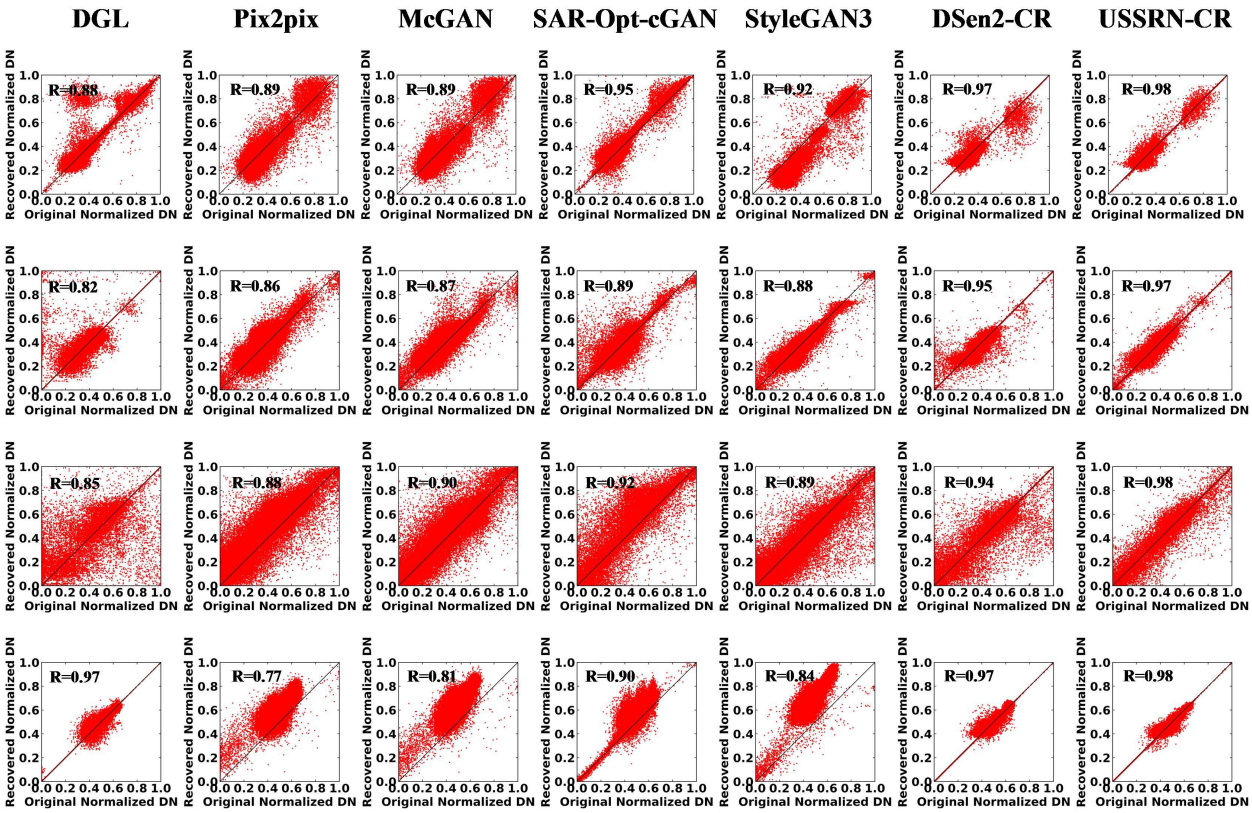


Fig. 10. Scatter diagrams between the original and reconstructed pixels in Fig. 9.

TABLE IV
QUANTITATIVE EVALUATION ON AIRBORNE DATASET

Method	PSNR↑	SSIM↑	CC↑	SAM↓
DGL	19.902	0.727	0.861	3.037
Pix2pix	18.336	0.478	0.836	8.806
McGAN	18.909	0.632	0.861	8.356
SAR-Opt-cGAN	19.538	0.706	0.874	5.534
StyleGAN3	20.856	0.570	0.890	7.108
DSen2-CR	21.926	0.771	0.917	2.637
USSRN-CR	**24.560**	**0.850**	**0.954**	**2.493**

observed that this model outperforms the other models in the reconstruction of tiny features. And the reconstructed image has clear edges and maintains consistency and continuity in structure. The DGL model performed well in reconstructing the spectrum of missing areas, but some details, such as continuous linear structure, are not perfect. The Pix2pix model and the McGAN model produce blurred boundary results, but the generated images have large spectral deviations and large changes in cloud-free areas. The StyleGAN3 model has a poor reconstruction effect on buildings and lower structural similarity to ground truth. The results generated by SAR-Opt-cGAN have obvious artifacts near damaged areas and the DSen2-CR model cannot recover the complete ground object boundary information, but both models maintain the image tone and texture details in cloud-free areas. Our model can accurately recover the structural information of farmlands, roads, and other ground objects, and the reconstruction of spectral information is also better than other algorithms.

Overall, compared with the other five algorithms, the model shows its superiority in the construction of structural texture information and the recovery of spectral information.

D. Real Experiment

For the real dataset, the SEN12MS-CR dataset is cropped into 256 pixels × 256 pixels. The sentinel-2 cloudy optic images are within 20% and 70% of cloud cover. As the cloud mask is needed in training. We use a combination of the methods proposed in [69] and [70] for cloud and cloud-shadow detection. The setting of parameters for cloud and cloud shadow detection is consistent with [37]. The threshold value of cloud binarization is $T_{\mathrm{CL}} = 0.2$, and the value greater than 0.2 is considered as a cloud. The threshold of cloud shadow detection is calculated according to the parameters $T_{\mathrm{CSI}} = 3/4$ and $T_{\mathrm{WBI}} = 5/6$.

In order to verify the practicability of our model on real data, we carried out real experiments based on the SEN12MS-CR dataset. We also use the sentinel-2 cloud images and SAR images for training, and present cloud-free sentinel-2 optical images from another time as the ground truth to compare quantitatively with other algorithms. The three images are acquired within the same meteorological season and we also

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

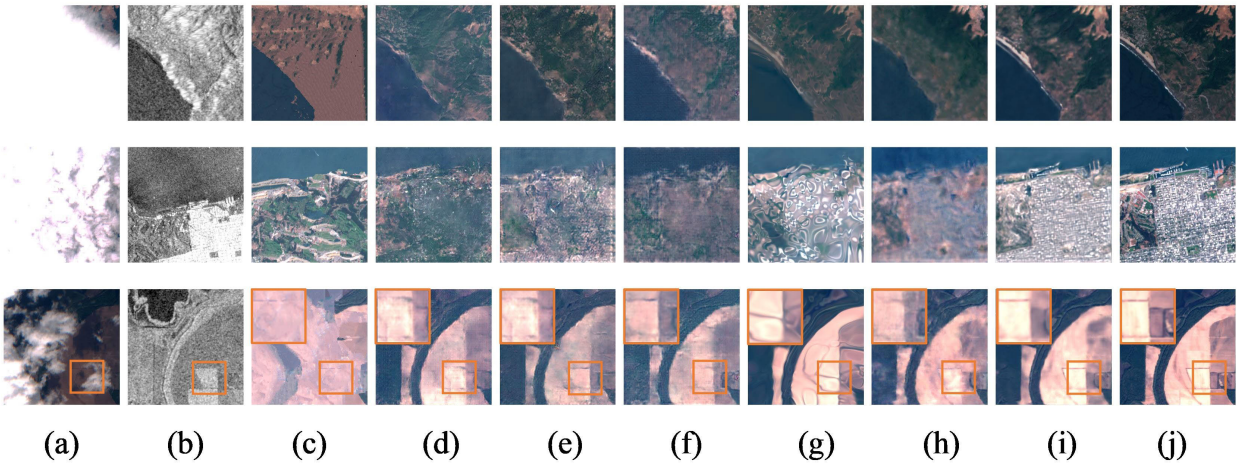


Fig. 11. Experimental results on real dataset. (a) Corrupted images. (b) SAR images. (c) Results of the DGL model. (d) Results of the Pix2pix model. (e) Results of the McGAN model. (f) Results of the SAR-Opt-cGAN model. (g) Results of the StyleGAN3 model. (h) Results of the DSen2-CR model. (i) Results of the USSRN-CR model. (j) Ground truth.

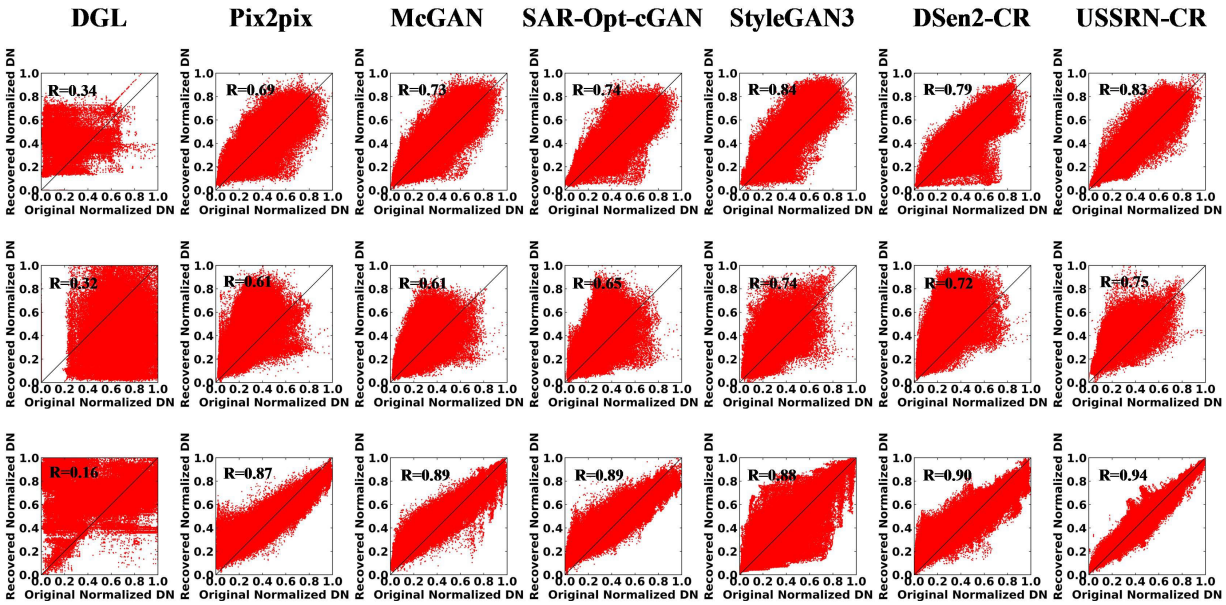


Fig. 12. Scatter diagrams between the original and reconstructed pixels in Fig. 10.

remove the images with large differences to limit surface changes.

The experiment results are displayed in Fig. 11. Fig. 11(a) and (b) shows, respectively, the cloud images and the SAR image. The results of the Pix2pix model, the McGAN model, the SAR-Opt-cGAN model, the StyleGAN3 model, the DSen2-CR model, and the USSRN-CR model are displayed in Fig. 11(c)–(i). Fig. 11(j) presents the real-ground image data. In addition, Fig. 12 shows the scatter diagrams of the real images and their R^2 which verify the superiority of the proposed model.

In the real dataset, the Pix2pix and McGAN models still have large spectral biases. In contrast, the proposed model achieves better visual results in the reconstructed regions and shows better performance than other algorithms in the quantitative evaluation in Table V. Fig. 11 shows the reconstruction results of different land cover, including cities, rivers, mountains, and farmlands. Our model shows the best restoration performance in terms of color consistency, texture consistency, and detail preservation. In the second scene,

TABLE V
QUANTITATIVE EVALUATION ON THE SEN12MS-CR DATASET

Method	PSNR↑	SSIM↑	CC↑	SAM↓
DGL	18.782	0.145	0.296	17.454
Pix2pix	35.161	0.843	0.579	6.332
McGAN	36.502	0.879	0.608	5.924
SAR-Opt-cGAN	36.742	0.900	0.621	5.568
StyleGAN3	38.699	0.845	0.664	4.805
DSen2-CR	37.862	0.909	0.629	4.910
USSRN-CR	**40.173**	**0.933**	**0.708**	**3.999**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

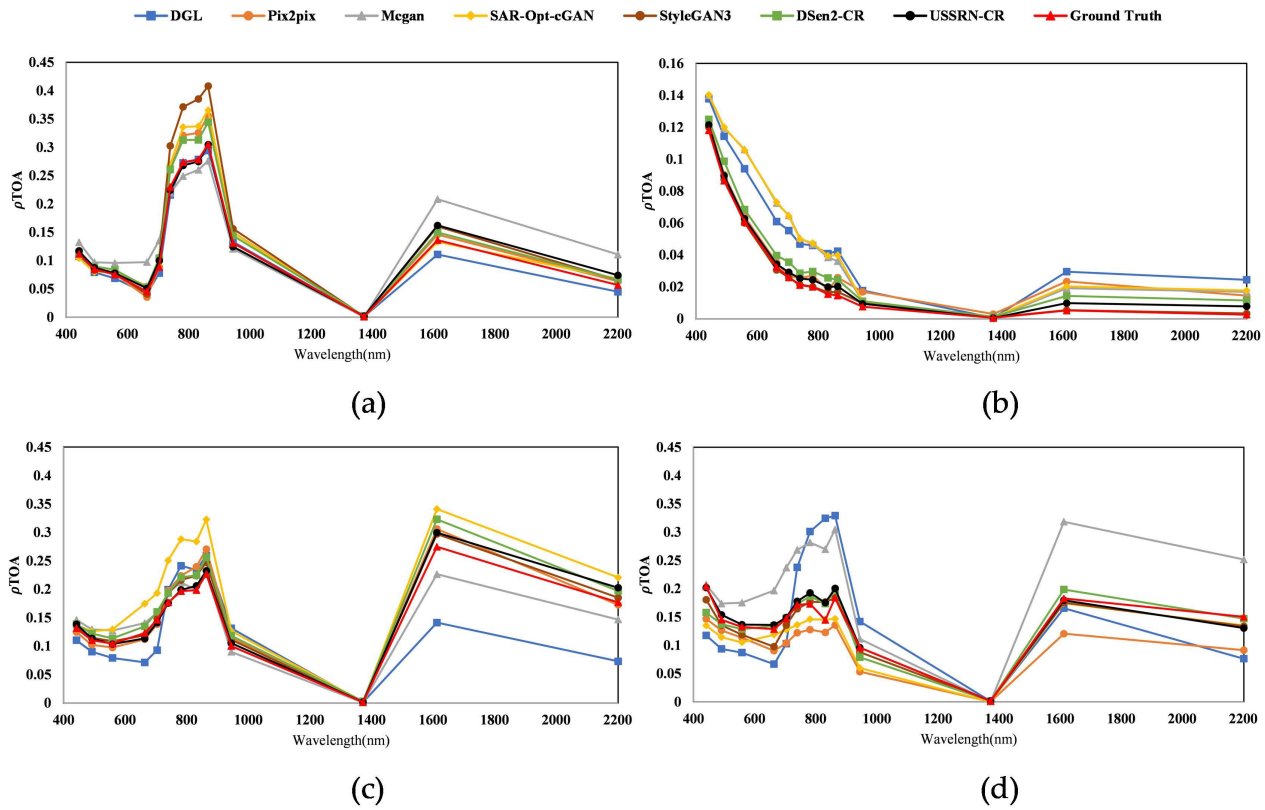


Fig. 13. Spectral curves of different feature types. (a) Vegetation. (b) Water. (c) Bare soil. (d) Artificial.

another model fails to reconstruct urban areas. The DSen2-CR model can only reconstruct the overall structure of the urban area in the case of large areas covered by thick clouds, while our model can reconstruct more details of the tiny features in the urban area and achieve better visual effects.

In order to evaluate the quantitative application ability of the repaired data, we selected various typical objects, including vegetation, bare soil, artificial, and water, to compare and analyze the fidelity of spectral and radiation characteristics based on the reconstruction results. The spectral curves of the top-of-atmosphere reflectance ρTOA based on various comparison methods are also shown in Fig. 13. Except that the short-wave infrared spectrum of vegetation is slightly weaker than that of the DSen2-CR model, the spectral curve of our method is the closest to the ground truth; especially, the high near-infrared reflection is almost the same. For water and bare soil objects, the DGL model, the McGAN model, and the SAR-Opt-cGAN model show large spectral deviation, while our model and the StyleGAN3 model are similar to the ground truth. Also, for artificial objects, the reconstructed results of another model also differ from the ground truth. However, the overall spectral curves of our model are very consistent except for some differences in a few spectral segments, such as 842 nm. In general, our model can restore spectral and radiation characteristics accurately.

Besides, the classification results for the DGL model, the Pix2pix model, the McGAN model, the SAR-Opt-cGAN model, the StyleGAN3 model, the DSen2-CR model, and the proposed model were produced to further verify the effectiveness of the results. Also, we classified cloud-free images at another time as ground truth to evaluate the accuracy of the reconstructed results. The land-cover classification results of cloud areas with a size of 1600 pixels $\times$ 1600 pixels were acquired with the maximum likelihood supervised classifier are presented in Fig. 14. The classification elements comprised the following categories: vegetation, bare soil, artificial, and water. Table VI lists the overall accuracy (OA) and kappa coefficient values. For the proposed model, the values of OA and kappa are 93.0% and 0.895, respectively, which are both superior to other algorithms. The classification results are similar and clear to the ground truth. In comparison, the bare soil was mostly classified as vegetation in the classification results of the DGL model, the Pix2pix model, and the McGAN model. The artificial classes of the DSen2-CR model and the SAR-Opt-cGAN model have poor accuracies and present more misclassifications. The bare soil classes by the StyleGAN3 model are fragmentary. This also demonstrates the superiority of the proposed model for classification.

In conclusion, the DGL model can achieve a better reconstruction effect when the land cover types are relatively simple. However, when there is a large area of thick cloud cover or complex land cover types, the reconstructed results display lower similarity to the cloud-free regions. Compared with the Pix2pix model and the McGAN model, which directly converts SAR images to optical images, the addition of cloud-based optical images makes the restoration results show better

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

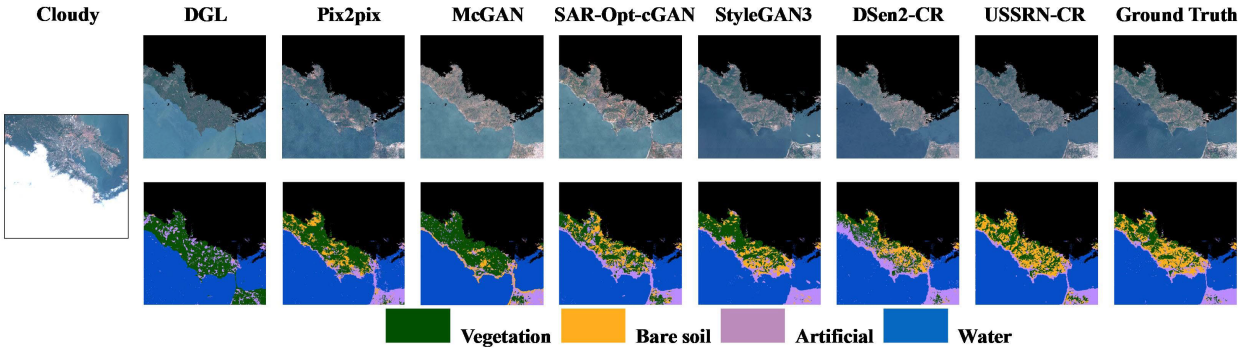


Fig. 14. Classification results of the different algorithms in the cloud and cloud shadow-covered regions.

TABLE VI
QUANTITATIVE CLASSIFICATION INDEXES (OA/KAPPA)

Data	DGL	Pix2pix	McGAN	SAR-Opt-cGAN	StyleGAN3	DSen2-CR	USSRN-CR
Sentinel-2	85.5/0.783	90.9/0.864	89.3/0.839	92.3/0.885	92.4/0.887	91.0/0.865	**93.0/0.895**

performance in spectral information reconstruction. Therefore, images reconstructed by the SAR-Opt-cGAN model, the StyleGAN3 model, the DSen2-CR model, and our model have better color consistency. However, the SAR-Opt-cGAN model, the StyleGAN3 model, and the DSen2-CR model have poor reconstruction effects on the texture details of ground objects and cannot maintain boundary consistency. Our model outperforms the other five contrasting algorithms in terms of visual effects, spectral fidelity, and reconstruction of texture details. Even under large areas of thick clouds, the model also achieves good reconstruction results. The simulation and real experiments of different spatial scales, different remote sensors, and different land cover also proved the applicability and robustness of the model.

TABLE VII
QUANTITATIVE EVALUATION ON ABLATION STUDY OF THE MODEL AND WITHOUT USE OF MFBS, CANNY OPERATOR (CANNY), THE GC, SCAB, AND SAR

Method	PSNR↑	SSIM↑	CC↑	SAM↓
Model w/o MFB	23.865	0.836	0.945	2.557
Model w/o CANNY	24.082	0.839	0.948	2.523
Model w/o GC	23.918	0.834	0.946	2.541
Model w/o SCAB	23.816	0.834	0.945	2.527
Model w/o SAR	22.615	0.804	0.928	2.734
USSRN-CR	**24.560**	**0.850**	**0.954**	**2.493**

E. Further Discussion

As described in Section III, our cloud removal model is characterized by several new blocks that were not utilized in previous SAR-based cloud removal studies. In this section, we analyze the effectiveness of each structure by comparing a few variants with and without the use of multiscale features blocks (MFBs), canny operator (CANNY), GC, SCAB, and SAR image (SAR) on the airborne dataset to evaluate the importance of these operations. The qualitative and quantitative evaluation results of these ablation models and the proposed model are presented in Table VII and Fig. 15. Besides, parameter sensitivity analysis and computational cost are also included in this section.

1) Ablation Study of the MFB: The MFB can obtain features with multicontext information to reconstruct ground objects of various scales. We train our model by reserving the 3×3 convolution layer but not using the MFB, denoted as w/o MFB. Table VII lists the quantitative evaluation results, including PSNR, SSIM, CC, and SAM. When the module for receptive field expanding and multiscale feature fusion was introduced, the model acquired high-accuracy results, which have better spatial continuity and showed a better performance in the quantitative assessment which demonstrated that the module seemed to be effective in cloud removal.

2) Ablation Study of the CANNY: The canny operator can extract the edge information of the SAR image for subsequent restoration. Compared with w/o CANNY, it can be observed that our model with the canny algorithm managed to restore more line structural details, such as the roads, and obtain better visual effects. The PSNR of USSRN-CR was 0.47 higher than the model without the canny operation and other indicators improved, which indicated that the module is effective.

3) Ablation Study of the GC: Compared with VC, GC can distinguish the valued pixels and nonvalued pixels and make full use of information in cloud-free areas to extract features. Model w/o GC indicates that ordinary convolution replaces the gated convolution in our model. As shown in the third line of Fig. 15, the results of the model with GC are clearer in the recovered building regions due to the better detail-preserving ability. Also, after introducing the GC, all of the indicator scores improved, especially the PSNR score, which increased by 0.65. The qualitative and quantitative results show that the details and textures blurred by VC are repaired by the GC.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

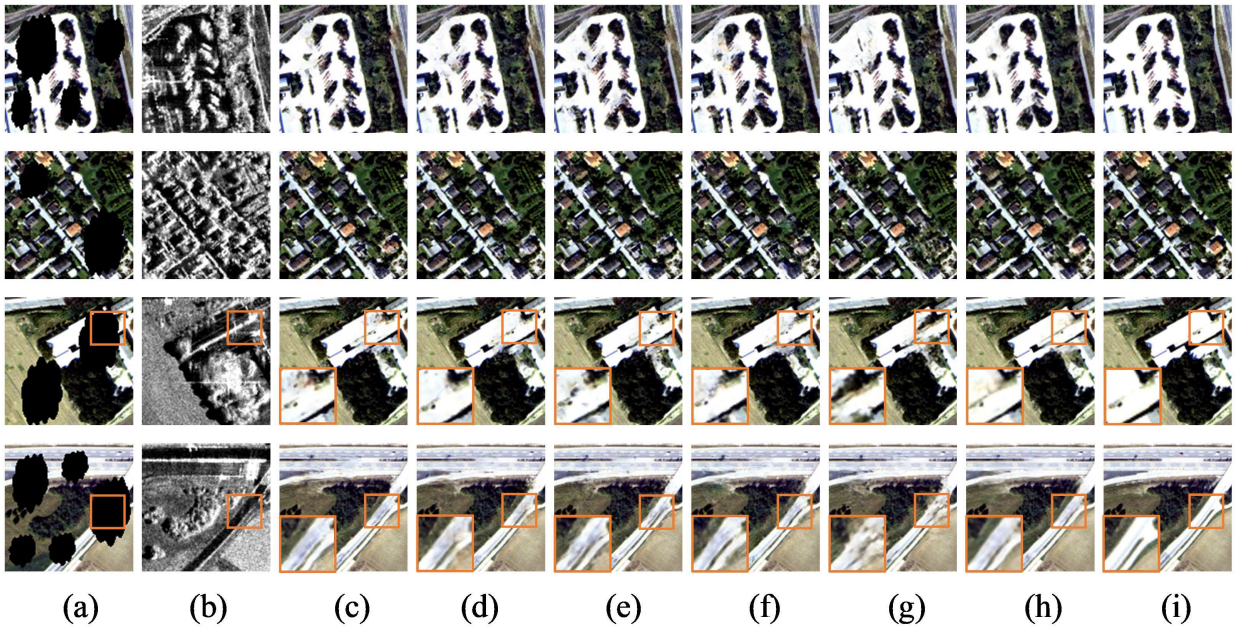


Fig. 15. Qualitative ablation study by different USSRN-CR networks. (a) Cloudy images. (b) SAR images. (c) Results without MFB. (d) Results without CANNY. (e) Results without GC. (f) Results without SCAB. (g) Results without SAR images. (h) Results with USSRN-CR. (i) Ground truth.

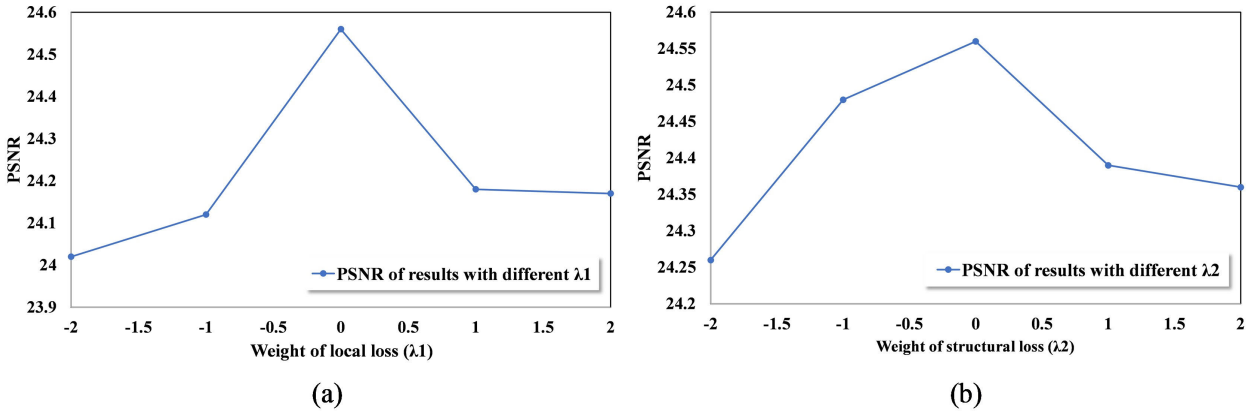


Fig. 16. Results of the proposed model with different weights. (a) λ_1. (b) λ_2.

4) Ablation Study of the SCAB: The SCAB is based on the attention mechanism, which introduces nonneighborhood spatial information and channel information as auxiliary information for cloud removal. Capturing global features plays a vital role in maintaining a global consistent structure. We train the USSRN-CR network by removing the SCAB, denoted as w/o SCAB. In the second and fourth scenes of Fig. 15, the recovery results of our full model are superior to the model without SCAB in terms of spectral fidelity of buildings and road texture. As previously stated, the quantitative evaluation is displayed in Table VII. The PSNR score increased by 0.74 after introducing the SCAB, which proved that the module can improve the performance of a cloud removal network.

5) Ablation Study of the SAR Data: The experiments were dedicated to verifying the usefulness of the SAR data. In Fig. 15, the results obtained by the model with and without SAR data input are visually compared. It can be seen that the results, which make use of SAR data contain much more structure information and can reconstruct more texture detail. In the first row of Fig. 15, the model without SAR data did not reconstruct the correct building. However, SAR images can provide the structure and texture of a building to assist in reconstructing the correct features in missing areas. The results using SAR optical data fusion contain more information than the results relying on optical image conversion, resulting in a greater improvement in the index. Compared with other modules, the PSNR indicator increased by 1.95. Therefore, SAR images of the same scene as prior data are beneficial for the overall cloud removal task.

6) Parameter Sensitive Analysis: We mainly analyze the parameters λ_1 and λ_2 in the loss function and fusion ratio of the optical image, SAR image, and the edge extracted from the canny operator, in order to fine-tune the model to the best state.

For the determination of the loss function, the weight of local loss λ_1 and the weight of structural loss λ_2 are set. There may be coupling between different loss functions,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUANTITATIVE EVALUATION ON ABLATION STUDY OF THE MODEL

Fusion Ratio	PSNR↑	SSIM↑	CC↑	SAM↓
1:1:1	24.142	0.843	0.952	2.534
2:1:1	24.419	0.847	0.953	2.516
3:1:1	24.373	0.848	0.953	2.520
4:1:1	24.369	0.848	0.953	2.565
2:1:2	24.438	0.848	0.952	2.518
2:2:1	**24.560**	**0.850**	**0.954**	**2.493**

TABLE IX
FLOPS, PARAMS, AND TIME OF DIFFERENT ALGORITHM ON AIRBORNE DATASET

Method	FLOPs(G)	Params(M)	Time(s)
DGL	\	\	21600.00
Pix2pix	4.51	54.40	3.13
McGAN	4.51	54.40	4.19
SAR-Opt-cGAN	4.53	54.41	4.94
StyleGAN3	33.12	28.13	18.62
DSen2-CR	310.79	18.90	32.11
USSRN-CR	305.80	18.61	32.88

so experiments are carried out based on the strategy of adding loss terms one by one. To quantitatively evaluate the results, PSNR was applied to compare models with different weights. As shown in Fig. 16(a), we first experimentally fix the weight of λ_1 to 1 at which point the PSNR reaches its maximum value. As λ_2 varies from 10^{-2} to 10^2, which is plotted in Fig. 16(b), the model achieves the highest PSNR when λ_2 equals 1. Therefore, we set λ_2 to 1.

For the fusion ratio, we carried out experiments based on different ratios and determined the parameter values one by one. The quantification indicators of different scale images are shown in the table VIII. First, the scale is fixed of the optical image through experiments, and the scale of the SAR image and the edge extracted by the canny operator is adjusted to determine the final fusion scale. The scale of optic images varies from 1 to 4 and the proportion of cloudy optical images gradually increases, the optimal index is reached when the scale is 2. Next, we carry out an experiment on the SAR image ratio and the edge ratio. We finally choose an optimal ratio of 2:2:1 for the optical image, SAR image, and edge map, which shows an advantage in quantitative indexes.

Relatively speaking, both optical images and SAR images contain more information on ground objects, and the edge map has less information, so the ratio of the optical image and SAR image is the same and more than that of the edge map. At this time, the network obtains the best results.

7) Computational Cost: Table IX lists the FLOPs, params, and time of the proposed model and the comparison algorithm on the airborne datasets. It can be seen that our model has fewer parameters than the other models. Due to the introduction of modules, such as gated convolution and attention mechanism, which are more complex in calculation and time-consuming, our model has higher FLOPs and computation time and is similar to the DSen2-CR model. For a simulated test set image of size 128 pixels × 128 pixels, other deep learning algorithms take only a few seconds, while the StyleGAN3 model, the DSen2-CR model, and our model take tens of seconds, and longer for the DGL model. Overall, the running time of the proposed model is acceptable, and superior qualitative and quantitative evaluation results are obtained, which attains a better balance between efficiency and effectiveness.

V. CONCLUSION

In this article, we use SAR data as auxiliary reference data to propose a unified spatial–spectral residual network for cloud removal from optical remote sensing images. In contrast to existing deep learning methods, we use gated convolution to replace ordinary convolution to realize the identification of valid pixels and invalid pixels. Also, the spatial and channel attention mechanisms are added to realize the utilization of global information and channel information. Simulated and real experimental results on GF-2/3, airborne SAR/optical data, and sentinel-1/2 confirm the feasibility and effectiveness of our model and its superiority over other deep-learning-based methods.

Although the proposed method has achieved satisfactory results in both qualitative and quantitative assessments, there are still some limitations that need to be overcome. Considering the difference between the cloudy images of the real dataset and the cloud-free images at a similar time, we plan to prepare a simulated dataset of large areas. At this time, it is necessary to ensure that the acquisition time of SAR images and cloud-free optical images is close. At present, there are a large number of spaceborne SAR satellites, and the revisit period is short. For example, GF-3 satellites can repeat transit within 1.5 days, which provides sufficient data for image restoration and ensures that the acquisition time of SAR data and optical data to be repaired is close to each other, thus reducing the error caused by time differences. Also, the category image or multiangle SAR data as auxiliary information will be considered when subsequent high-resolution image restoration.

REFERENCES

[1] M. A. Wulder et al., "The global Landsat archive: Status, consolidation, and direction," *Remote Sens. Environ.*, vol. 185, pp. 271–283, Nov. 2016.
[2] W. B. Rossow, A. W. Walker, and L. C. Garder, "Comparison of ISCCP and other cloud amounts," *J. Climate*, vol. 6, no. 12, pp. 2394–2418, Dec. 1993.
[3] H. Shen et al., "Missing information reconstruction of remote sensing data: A technical review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 61–85, Sep. 2015.
[4] C. Zhang, W. Li, and D. Travis, "Gaps-fill of SLC-off Landsat ETM+ satellite image using a geostatistical approach," *Int. J. Remote Sens.*, vol. 28, no. 22, pp. 5103–5122, Nov. 2007.
[5] R. E. Rossi, J. L. Dungan, and L. R. Beck, "Kriging in the shadows: Geostatistical interpolation for remote sensing," *Remote Sens. Environ.*, vol. 49, no. 1, pp. 32–40, Jul. 1994.
[6] M. Bertalmio, L. Vese, G. Sapiro, and S. Osher, "Simultaneous structure and texture image inpainting," *IEEE Trans. Image Process.*, vol. 12, no. 8, pp. 882–889, Aug. 2003.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

[7] R. Mendez-Rial, M. Calvino-Cancela, and J. Martin-Herrero, "Anisotropic inpainting of the hypercube," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 214–218, Mar. 2012.

[8] A. Criminisi, P. Perez, and K. Toyama, "Region filling and object removal by exemplar-based image inpainting," *IEEE Trans. Image Process.*, vol. 13, no. 9, pp. 1200–1212, Sep. 2004.

[9] K. He and J. Sun, "Image completion approaches using the statistics of similar patches," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 12, pp. 2423–2435, Dec. 2014.

[10] J. Shen and T. F. Chan, "Mathematical models for local nontexture inpaintings," *SIAM J. Appl. Math.*, vol. 62, no. 3, pp. 1019–1043, Jan. 2002.

[11] R. C. Hardie, K. J. Barnard, and E. E. Armstrong, "Joint MAP registration and high-resolution image estimation using a sequence of undersampled images," *IEEE Trans. Image Process.*, vol. 6, no. 12, pp. 1621–1633, Dec. 1997.

[12] Q. Cheng, H. Shen, L. Zhang, and P. Li, "Inpainting for remotely sensed images with a multichannel nonlocal total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 175–187, Jan. 2014.

[13] M. K. Ng, Q. Yuan, L. Yan, and J. Sun, "An adaptive weighted tensor completion method for the recovery of remote sensing images with missing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3367–3381, Jun. 2017.

[14] J. Chen, X. Zhu, J. E. Vogelmann, F. Gao, and S. Jin, "A simple and effective method for filling gaps in Landsat ETM+ SLC-off images," *Remote Sens. Environ.*, vol. 115, no. 4, pp. 1053–1064, Apr. 2011.

[15] X. Zhu, F. Gao, D. Liu, and J. Chen, "A modified neighborhood similar pixel interpolator approach for removing thick clouds in Landsat images," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 521–525, May 2012.

[16] B. Chen, B. Huang, L. Chen, and B. Xu, "Spatially and temporally weighted regression: A novel method to produce continuous cloud-free Landsat imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 27–37, Jan. 2017.

[17] Z. Li, H. Shen, Q. Cheng, W. Li, and L. Zhang, "Thick cloud removal in high-resolution satellite images using stepwise radiometric adjustment and residual correction," *Remote Sens.*, vol. 11, no. 16, p. 1925, Aug. 2019.

[18] H. Shen, J. Wu, Q. Cheng, M. Aihemaiti, C. Zhang, and Z. Li, "A spatiotemporal fusion based cloud removal method for remote sensing images with land cover changes," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 862–874, Mar. 2019.

[19] X. Li, L. Wang, Q. Cheng, P. Wu, W. Gan, and L. Fang, "Cloud removal in remote sensing images using nonnegative matrix factorization and error correction," *ISPRS J. Photogramm. Remote Sens.*, vol. 148, pp. 103–113, Feb. 2019.

[20] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[21] R. Eckardt, C. Berger, C. Thiel, and C. Schmullius, "Removal of optically thick clouds from multi-spectral satellite images using multi-frequency SAR data," *Remote Sens.*, vol. 5, no. 6, pp. 2973–3006, Jun. 2013.

[22] G. Xingfa and T. Xudong, "Overview of China earth observation satellite programs [space agencies]," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 113–129, Sep. 2015.

[23] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[24] C. Zhu, Z. Zhao, X. Zhu, Z. Nie, and Q. H. Liu, "Cloud removal for optical images using SAR structure data," in *Proc. IEEE 13th Int. Conf. Signal Process. (ICSP)*, Nov. 2016, pp. 1872–1875.

[25] L. Liu and B. Lei, "Can SAR images and optical images transfer with each other?" in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 7019–7022.

[26] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[27] J. D. Bermudez, P. N. Happ, D. A. B. Oliveira, and R. Q. Feitosa, "Sar to optical image synthesis for cloud removal with generative adversarial networks," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 1, pp. 5–11, Sep. 2018.

[28] J. D. Bermudez, P. N. Happ, R. Q. Feitosa, and D. A. B. Oliveira, "Synthesis of multispectral optical images from SAR/optical multitemporal data using conditional generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1220–1224, Aug. 2019.

[29] M. Mirza and S. Osindero, "Conditional generative adversarial nets," in *Proc. Neural Inf. Process. Syst. Deep Learn. Represent. Learn. Workshop*, Dec. 2014, pp. 2672–2680.

[30] C. Grohnfeldt, M. Schmitt, and X. Zhu, "A conditional generative adversarial network to fuse SAR and multispectral optical data for cloud removal from Sentinel-2 images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1726–1729.

[31] F. N. Darbaghshahi, M. R. Mohammadi, and M. Soryani, "Cloud removal in remote sensing images using generative adversarial networks and SAR-to-optical image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–9, Oct. 2021, Art. no. 4105309.

[32] Y. Li, R. Fu, X. Meng, W. Jin, and F. Shao, "A SAR-to-optical image translation method based on conditional generation adversarial network (cGAN)," *IEEE Access*, vol. 8, pp. 60338–60343, 2020.

[33] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nov. 2017, pp. 5967–5976.

[34] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4396–4405.

[35] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of StyleGAN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8107–8116.

[36] T. Karras et al., "Alias-free generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, Jun. 2021, pp. 852–863.

[37] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, Aug. 2020.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[39] Q. Zhang, Q. Yuan, J. Li, Z. Li, H. Shen, and L. Zhang, "Thick cloud and cloud shadow removal in multitemporal imagery using progressively spatio-temporal patch group deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 148–160, Apr. 2020.

[40] J. Gao, Y. Yi, T. Wei, and G. Zhang, "Sentinel-2 cloud removal considering ground changes by fusing multitemporal SAR and optical images," *Remote Sens.*, vol. 13, no. 19, p. 3998, Oct. 2021.

[41] H. Chen, N. Yokoya, C. Wu, and B. Du, "Unsupervised multimodal change detection based on structural relationship graph representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, Dec. 2022, Art. no. 5635318.

[42] Y. Bengio, P. Simard, and P. Frasconi, "Learning long-term dependencies with gradient descent is difficult," *IEEE Trans. Neural Netw.*, vol. 5, no. 2, pp. 157–166, Mar. 1994.

[43] F. Xu et al., "GLF-CR: SAR-enhanced cloud removal with global–local fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 268–278, Oct. 2022.

[44] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, and B. Catanzaro, "Image inpainting for irregular holes using partial convolutions," in *Proc. Eur. Conf. Comput. Vis.*, Jun. 2018, pp. 85–100.

[45] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. Huang, "Free-form image inpainting with gated convolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4470–4479.

[46] P. Dai, S. Ji, and Y. Zhang, "Gated convolutional networks for cloud removal from bi-temporal remote sensing images," *Remote Sens.*, vol. 12, no. 20, p. 3427, Oct. 2020.

[47] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2014, pp. 818–833.

[48] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, Dec. 2017, pp. 1–11.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.

[49] H. Luo, C. Chen, L. Fang, X. Zhu, and L. Lu, "High-resolution aerial images semantic segmentation using deep fully convolutional network with channel attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3492–3507, Sep. 2019.

[50] S. Zhu, J. Zhang, X. Liang, and Q. Guo, "Multiscale semantic guidance network for object detection in VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[51] L. Ru, B. Du, and C. Wu, "Multi-temporal scene classification and scene change detection with correlation based fusion," *IEEE Trans. Image Process.*, vol. 30, pp. 1382–1394, 2021.

[52] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Generative image inpainting with contextual attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5505–5514.

[53] M. Xu, F. Deng, S. Jia, X. Jia, and A. J. Plaza, "Attention mechanism-based generative adversarial networks for cloud removal in Landsat images," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112902.

[54] S. Zhuoran, Z. Mingyuan, Z. Haiyu, Y. Shuai, and L. Hongsheng, "Efficient attention: Attention with linear complexities," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3530–3538.

[55] X. Wen, Z. Pan, Y. Hu, and J. Liu, "An effective network integrating residual learning and channel attention mechanism for thin cloud removal," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[56] H. Liu, B. Jiang, Y. Xiao, and C. Yang, "Coherent semantic attention for image inpainting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4169–4178.

[57] J. Li, N. Wang, L. Zhang, B. Du, and D. Tao, "Recurrent feature reasoning for image inpainting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7757–7765.

[58] W. Li, Y. Li, and J. C. Chan, "Thick cloud removal with optical and SAR imagery via convolutional-mapping-deconvolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2865–2879, Apr. 2020.

[59] K. Nazeri, E. Ng, T. Joseph, F. Qureshi, and M. Ebrahimi, "EdgeConnect: Structure guided image inpainting using edge prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3265–3274.

[60] J. Li, F. He, L. Zhang, B. Du, and D. Tao, "Progressive reconstruction of visual structure for image inpainting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5961–5970.

[61] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jun. 2010, pp. 807–814.

[62] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, May 2019, pp. 7354–7363.

[63] J. Gao, Q. Yuan, J. Li, H. Zhang, and X. Su, "Cloud removal with fusion of high resolution optical and SAR images using generative adversarial networks," *Remote Sens.*, vol. 12, no. 1, p. 191, Jan. 2020.

[64] O. Hellwich, A. Reigber, and H. Lehmann, "Sensor and data fusion contest: Test imagery to compare and combine airborne SAR and optical sensors for mapping," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2002, pp. 82–84.

[65] S. Parrilli, M. Poderico, C. V. Angelino, and L. Verdoliva, "A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 606–616, Feb. 2012.

[66] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, Dec. 2014, pp. 1–15.

[67] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[68] F. A. Kruse et al., "The spectral image processing system (SIPS)-interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 145–163, May 1993.

[69] M. Schmitt, L. Hughes, C. Qiu, and X. X. Zhu, "Aggregating cloud-free Sentinel-2 images with Google Earth engine," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, pp. 145–152, Sep. 2019.

[70] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Cloud/shadow detection based on spectral indices for multi/hyperspectral optical remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 235–253, Oct. 2018.



Yuxi Wang received the B.S. degree from the School of Geography and Information Engineering, Lanzhou University, Lanzhou, China, in 2020. She is currently pursuing the Ph.D. degree in cartography and geographic information systems with the Key Laboratory of Computational Optical Imaging Technology and the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

Her research interests include feature extraction, optical image inpainting, multimodal remote sensing data fusion, and deep learning.



Bing Zhang (Fellow, IEEE) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS, where he has been leading lots of key scientific projects in the area of hyperspectral remote sensing for more than 25 years. He has edited six books/contributed book chapters on hyperspectral image processing and subsequent applications. He has developed five software systems in image processing and applications. His creative achievements received ten important prizes from the Chinese government and special government allowances from the Chinese State Council. He has authored more than 300 publications, including more than 170 journal articles. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang has been serving as a Technical Committee Member for the IEEE Workshop on Hyperspectral Image and Signal Processing since 2011, the President for the Hyperspectral Remote Sensing Committee of the China National Committee of the International Society for Digital Earth since 2012, and the Standing Director for the Chinese Society of Space Research since 2016. He is a Student Paper Competition Committee Member of the International Geoscience and Remote Sensing Symposium from 2015 to 2019. He was a recipient of the National Science Foundation for Distinguished Young Scholars of China Award in 2013 and the 2016 Outstanding Science and Technology Achievement Prize of the Chinese Academy of Sciences, the highest level of awards for CAS scholars. He is currently serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Wenjuan Zhang (Member, IEEE) received the B.S. degree in land resource management from Wuhan University, Wuhan, China, in 2003, and the Ph.D. degree in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2008.

She is currently a Professor with the Airborne Remote Sensing Center, Aerospace Information Research Institute, CAS. Her research interests include remote sensing image quality improvement, image simulation, and payload characterize analysis.

Dr. Zhang received the Outstanding Science and Technology Achievement Prize of CAS in 2016 and the Second Prize of the State Scientific and Technological Progress Award in 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.



Danfeng Hong (Senior Member, IEEE) received the M.Sc. degree (summa cum laude) in computer vision from the College of Information Engineering, Qingdao University, Qingdao, China, in 2015, and the Dr.-Ing. degree (summa cum laude) from the Signal Processing in Earth Observation, Technical University of Munich, Munich, Germany, in 2019.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. He was a Research Scientist and a Leader of the Spectral Vision Working Group, Remote Sensing Technology Institute, German Aerospace Center, Oberpfaffenhofen, Germany. He was also an Adjunct Scientist with the Grenoble Images Parole Signal et Automatique - Laboratory (GIPSA-Lab), the Grenoble Institute of Technology (Grenoble INP), Centre national de la recherche scientifique (CNRS), Université Grenoble Alpes, Grenoble, France. His research interests include artificial intelligence, remote sensing big data analysis, multimodal interpretation, and their applications in Earth vision.

Dr. Hong is an Editorial Board Member of *Remote Sensing* and an Editorial Advisory Board Member of the International Society of Photogrammetry and Remote Sensing (ISPRS). He was a recipient of the Best Reviewer Award of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2021 and 2022, the Best Reviewer Award of the IEEE *Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (JSTARS) in 2022, the Jose Bioucas Dias Award for recognizing the Outstanding Paper at WHISPERS in 2021, the Remote Sensing Young Investigator Award in 2022, the IEEE Geoscience and Remote Sensing Society (GRSS) Early Career Award in 2022, and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) in 2022. He is an Associate Editor of the IEEE TGRS.



Bin Zhao (Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2021.

From 2018 to 2023, he was a Scientist with the Faculty of Electrical and Computer Engineering, University of Iceland, under the supervision of Prof. Jón Atli Benediktsson, Prof. Magnus O. Ulfarsson, and Prof. Jocelyn Chanussot. He is currently a Professor and the Head of the Hyperspectral Remote Sensing Laboratory, School of Information Science and Engineering, Shandong Agricultural University, Tai'an, China. He received the Doctoral Grant of the University of Iceland Research Fund and the Icelandic Government Research Fund twice in 2018 and 2021, respectively. His research interests include hyperspectral image denoising, super-resolution, feature extraction, classification, and agricultural remote sensing.

Dr. Zhao was a recipient of the 2021 Chinese Government Award for the Outstanding Self-Financed Students Abroad. He is the first Chinese student in Iceland to receive the award. From all over the world, 650 young talents received the award from the China Scholarship Council. He has served as a Co-Chair for the IEEE International Geoscience and Remote Sensing Symposium in 2021. He is a Guest Editor of *Remote Sensing*.



Zhen Li received the B.Sc. degree in communication engineering from Northwestern Polytechnical University, Xi'an, China, in 2015, and the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2021.

He is currently an Assistant Research Fellow with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His current research interests include computer vision, machine learning, with an emphasis on hyperspectral image processing, remote sensing images, super-resolution, and quality assessment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:37:05 UTC from IEEE Xplore. Restrictions apply.